When Does Test-Time Ensembling Help Large Language Models? A Budget-Matched Study of Answer-Space and Prompt-Space Diversity

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Abstract

Test-time ensembling for large language models draws diversity from two sources: resampling the stochastic decoder on a fixed prompt (answer space) or aggregating semantically equivalent paraphrases (prompt space). We compare them budget-matched, answering every question sixteen times per model. A development suite (90 questions, six systems) fixed the conditions; the primary evidence is a held-out suite of 119 fresh questions fully covered by four models from two families (476 model–question pairs). Against the single-sample expectation S1, both six-vote ensembles improve out of sample: self-consistency by +2.4 points and the paraphrase ensemble by +4.9 (95% cluster-bootstrap intervals +1.1 to +3.7 and +2.3 to +7.6), with the twelve-vote combination at +4.1; the two sources remain statistically inseparable (+2.5, interval touching zero), and +1.9 points of the paraphrase headline is a single-call difficulty offset, leaving +3.0 of within-family aggregation against +2.4 for resampling. The margin is conditional: it concentrates where answer-space diversity is exhausted (computation-deterministic tasks), reverses where paraphrases distort verbal-reasoning stems, and its realizing allocation is model-dependent. On saturated benchmarks, answer distributions collapse, no strategy moves accuracy, and agreement-based uncertainty estimates lose their signal. Greedy divergence from S1 is endpoint-specific (three of ten configurations). An adaptive probe-and-route rule transfers at

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79–87 percent of the six-call budget, beating every fixed allocation on one model and trailing the best fixed conditions elsewhere by one to seven questions; a sequential unanimity rule preserves accuracy at two to three calls. We release prompts, paraphrases, per-call logs, and recomputation scripts.

Keywords: Large language models, test-time compute, self-consistency, prompt ensembling, uncertainty estimation, robustness

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1. Introduction

Large language models answer questions through a single forward pass whose output depends on two design choices that the user controls: the prompt that specifies the task, and the decoding configuration that maps model probabilities into text. Both choices turn out to be fragile. A long line of work on prompt sensitivity shows that semantically equivalent reformulations of the same question can change model answers by large margins [1, 2, 3, 4]. A parallel line of work on test-time compute shows that a single greedy decode is an unreliable summary of what the model knows, and that aggregating several decodes of the same prompt, as in self-consistency [5], improves reasoning accuracy at the cost of additional inference calls.

These two observations motivate two different ways of spending a fixed budget of model calls. The first spends the budget on the answer space: keep the prompt fixed, sample several decodes at nonzero temperature, and aggregate them by majority vote. The second spends the budget on the prompt space: keep decoding deterministic or lightly stochastic, present several semantically equivalent paraphrases of the task, and aggregate the resulting answers. Both strategies are used in practice, both appear in the evaluation literature under separate headings, and both consume the same scarce resource. Surprisingly, the literature offers little guidance on which source of diversity yields more accuracy per call, whether the two sources detect the same errors, and how the answer changes with model capability and task difficulty.

The question is becoming more urgent as model behavior shifts. Post-trained flagship models are increasingly confident: on many legacy benchmarks they answer near ceiling, and preliminary reports suggest that repeated sampling at moderate temperature often returns identical answers, which would render answer-space ensembling inert precisely where deployments are most common. At the same time, uncertainty estimation methods

built on sampling, such as semantic entropy [6, 7], implicitly assume that the sample distribution carries information about correctness. If that assumption fails for confident models, practitioners need to know before they build on it.

This paper presents, to our knowledge, the first budget-matched, within-question comparison of answer-space and prompt-space ensembling across multiple model families and task types. Every question in our protocol is answered sixteen times by every model under matched conditions: one greedy decode, six stochastic decodes of the original prompt, one stochastic decode of each of six validated paraphrases, and three additional second decodes that support a hybrid condition. From this single call matrix we derive self-consistency ensembles of any size up to six, paraphrase ensembles of the same sizes, a hybrid allocation, a combined twelve-vote ensemble, per-question consistency scores for both diversity sources, and fragility measures that record how often paraphrasing alone flips the majority answer. Three follow-up conditions then isolate the factors the matrix alone cannot separate: a greedy-paraphrase control that removes decoding stochasticity from the paraphrase side, a chain-of-thought variant that changes the answering mode, and a hard-task probe on MMLU-Pro; all three are reported in the appendices as descriptive controls. The evaluation runs in two phases: a *development suite* of 90 questions on six models, whose unevenly truncated coverage (12 to 90 questions per model) reflects the free-tier quotas of the providers, fixes the conditions, strata, and rules; and a *held-out suite* of 119 fresh questions, fully covered by four models from two providers and two families, which carries the primary evaluation. Because all conditions are derived from matched calls, comparisons between strategies hold the question, the model, and the total call budget constant by construction.

Our study addresses four research questions. First, under equal budgets, does spending calls on paraphrases buy more, equal, or less accuracy than spending them on resampling, and does the hybrid allocation dominate either pure strategy? Second, how brittle are model answers to paraphrasing per se, and where do paraphrase-induced flips concentrate? Third, can agreement across paraphrases serve as an error signal in the way that agreement across samples does, and how do the two signals compare when both are available? Fourth, how do all of these effects interact with model capability and task difficulty? The last question turns out to govern the first three: on the held-out suite, where the votes disagree, both sources deliver small reliable gains over the single-sample expectation (pooled +2.4 and +4.9 points, confidence

intervals excluding zero) yet remain statistically inseparable from each other, and where the prompt-space margin appears it concentrates exactly where resampling is exhausted — a conditional answer, not a universal winner.

The empirical picture that emerges has a clear regime structure. When a model finds a task easy, its answer distribution collapses: across our six benchmarks, flagship models return the same answer for essentially every sample and every paraphrase on saturated tasks, self-consistency and paraphrase ensembling both collapse onto both single-call baselines, and consistency scores carry no signal about which answers are wrong. When a task genuinely challenges a model, the votes disagree, and the interesting quantity stops being whether to aggregate and becomes what the aggregation is measured against: a single stochastic draw is already close to what six or twelve votes return on average, so the choice of the k=1 reference, greedy or single-sample, governs whether ensembling looks like a large gain or a wash, while the sign of the residual gain is set by whether the model's modal answer is correct. This structure implies that no single published claim about ensembling generalizes across regimes, and that practical systems should first measure the cheap statistics that identify the regime before choosing how to spend their budget.

Our contributions are as follows. We formalize test-time ensembling as a choice between two diversity sources under a shared budget, and we introduce a call matrix design that derives all conditions of interest from one matched set of calls, with controls that separate prompt-space effects from decoding stochasticity, answering mode, and task difficulty. We provide a multi-model, multi-benchmark empirical map of when each source helps, validated on a held-out suite of 119 fresh questions with four fully covered models, including systematic evidence that paraphrase agreement functions as an uncertainty signal whose error coverage only partially overlaps that of sample agreement. We document the answer degeneracy phenomenon that renders both ensembling and consistency-based uncertainty estimation inert on saturated tasks for confident models, with direct implications for sampling-based confidence estimation in this regime. Finally, we translate the findings into an adaptive allocation rule checked by simulation on the development matrices and transferred to held-out questions, and we release our complete prompt materials, validated paraphrases, per-call predictions, and the scripts that recompute every table from the raw logs.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the ensembling framework and the protocol.

Section 4 describes models, benchmarks, and validation procedures. Section 5 presents the development-suite results around our four research questions, followed by the held-out evaluation that carries the primary comparisons; three follow-up controls appear in Appendices Appendix A–Appendix C. Section 6 discusses implications and limitations, and Section 7 concludes.

2. Related Work

2.1. Test-time compute and self-consistency

Scaling computation at inference time has become a standard tool for improving the reliability of large language models. Chain-of-thought prompting demonstrates that eliciting intermediate reasoning steps raises accuracy on arithmetic and symbolic tasks [8], and self-consistency extends this observation by sampling multiple reasoning paths at nonzero temperature and returning the most frequent final answer [5]. Follow-up work reduces the number of samples needed through adaptive stopping criteria [9] and early-stopping rules that monitor the stability of the running majority [10], generalizes aggregation to free-form generation by letting the model itself select among candidate answers [11], and studies the optimal allocation of test-time compute relative to model size [12]. Repeated sampling studies document that coverage grows log-linearly with the sample budget but that majority voting plateaus without a verifier [13], a plateau our regime analysis explains: where the modal answer is wrong, additional votes amplify the error. Process reward models that score intermediate steps represent a further refinement of the same budget idea [14]. Multi-agent debate aggregates answers across multiple model instances and rounds [15], with prompt variation proposed as one axis of diversity of thought [16], and controlled comparisons find that debate does not reliably outperform self-consistency at matched budgets [17], which reinforces the need for budget-matched comparisons when evaluating aggregation strategies. Our work differs from this literature in the source of diversity it studies: instead of asking how many samples to draw or how to weight them, we ask whether the same calls are better spent perturbing the prompt, and we answer under a matched-budget protocol.

2.2. Prompt sensitivity and prompt ensembling

A separate literature documents that model outputs are sensitive to semantically irrelevant details of the prompt. Reformatting alone, such as changing spacing or option ordering, shifts accuracy by several points [2], and

the measured performance of a model depends strongly on which of many equivalent prompts an evaluator happens to choose [1]. Linguistic properties of the prompt, such as lexical familiarity, partially predict its success [3], and adversarial prompt perturbations can degrade performance further [4]. The natural constructive response is to aggregate over prompts. Ask Me Anything generates many question reformulations with a language model and combines the resulting pseudo-labels [18], an idea that predates and parallels self-consistency but targets zero-shot classification rather than matched-budget reasoning. Boosted prompt ensembles construct few-shot prompt sets stepwise on hard examples and outperform single-prompt output-space ensembles on GSM8K and AQuA [19]; diversity-of-thought prompting varies the prompt to induce diverse reasoning paths before aggregation [16]; and evaluation studies propose averaging over prompt templates to stabilize benchmark scores [1]. These methods share our premise that the prompt is a legitimate axis of ensembling, but none of them isolates prompt-space diversity from decoding stochasticity under a matched budget, and none quantifies when prompt-space aggregation beats answer-space aggregation on identical questions, which is the gap our protocol fills.

2.3. Uncertainty estimation from sampled behavior

Between the two aggregation literatures sits a third that uses sampling not to improve accuracy but to estimate uncertainty. Verbalized confidence asks the model to state a probability [20, 21], and self-probing methods ask the model to evaluate its own candidate answers [22]. Behavioral methods infer uncertainty from the dispersion of sampled outputs, from simple answer agreement to semantic clustering that treats paraphrased answers as equivalent [6, 7], with surveys consolidating the rapidly growing set of measures [23]. Perturbation-based methods broaden the probe beyond decoding: SPUQ perturbs the input, samples outputs per perturbation, and aggregates [24], and input clarification ensembling decomposes LLM uncertainty into data and model components by generating input clarifications and ensembling the predictions [25]. Calibration metrics such as expected calibration error [26] and selective prediction metrics such as the area under the risk-coverage curve quantify how well a confidence score ranks correct above incorrect answers. These methods inherit an assumption from sampling: that the sample distribution is informative. Our study tests this assumption directly, characterizes the regimes in which sample agreement and paraphrase agreement carry signal, and quantifies how much of the two signals' error

coverage overlaps. In this sense our fragility analysis also connects to work on the faithfulness of chain-of-thought explanations, which finds that stated reasoning does not always cause the final answer [27]: paraphrase stability measures the robustness of the answer to surface form rather than to the reasoning trace.

2.4. Positioning

In summary, answer-space ensembling, prompt-space ensembling, and sampling-based uncertainty estimation have developed largely in isolation. Answer-space methods assume resampling generates useful diversity. Prompt-space methods assume paraphrases probe the same knowledge. Uncertainty methods assume sample dispersion tracks correctness. All three assumptions are regime dependent, and no prior work, to our knowledge, measures all three under a single matched-budget protocol on modern models. The comparison also carries practical weight for evaluation practice, since benchmark conclusions about model capability shift when prompts vary [1] and when aggregation is applied [5]. Our contribution is the unified empirical map, the validation machinery for paraphrases, and the decision guidance that follows from the regime structure.

3. A Budget-Matched Framework for Test-Time Ensembling

3.1. Setup, notation, and suite roles

Let M denote a large language model and let q denote a question with a gold answer $y^*(q)$. For multiple-choice tasks the answer is an option letter, and for free-form numeric tasks the answer is a value compared numerically. A prompt x is a rendering of q into natural language, consisting of an optional passage, a question stem, an option list, and an instruction that specifies the required output format. We distinguish one canonical prompt x_0 and K paraphrased prompts $x_1, \ldots, x_K$ that preserve the semantics of x_0 under a validation procedure described in Section 4. The study uses two question suites in fixed roles: a *development suite* of 90 questions on which conditions, strata, and allocation rules were designed, and a *held-out suite* of 119 fresh questions, drawn mechanically from the same pools by replaying the seeded shuffle (Section 4) without consulting any model output, on which the primary evaluation is conducted. Because the strata and rules were fixed after seeing the development-suite results, the held-out suite is a transfer test of

previously chosen designs, not a preregistered confirmation; we use the term *held-out* rather than blind throughout.

The model produces an answer through stochastic decoding. We write $a(M, x, s)$ for the answer extracted from the s-th independent decode of prompt x at temperature $\tau = 0.7$, and $a_g(M, x)$ for the greedy decode at $\tau = 0$. Answers are extracted from the model output by a fixed parser that searches for the mandated answer format and falls back to pattern matching over the final output lines; the parse rate is reported in Section 4, together with a retry policy and a sensitivity analysis for failed extractions.

3.2. The call matrix

For every question and model we collect a matrix of sixteen calls:

$$\mathcal{C}(M, q) \;=\; \big\{a_g(M, x_0)\big\} \;\cup\; \bigcup_{s=1}^{S} \big\{a(M, x_0, s)\big\} \;\cup\; \bigcup_{k=1}^{K} \big\{a(M, x_k, s) : s \le r_k\big\}, \tag{1}$$

with $S = 6$ samples on the canonical prompt, $K = 6$ paraphrases, and $r_k = 2$ decodes for the first three paraphrases and $r_k = 1$ for the remaining three. The matrix therefore contains one greedy call, six canonical samples, six first-decode paraphrase calls, and three second-decode paraphrase calls. Every experimental condition studied in this paper is a deterministic function of this matrix, which guarantees that all comparisons hold the question, the model, and the realized decoding randomness constant across conditions.

3.3. Ensemble conditions

All conditions aggregate votes by majority rule. Ties are broken in favor of the earliest vote in a fixed canonical order, which prefers the canonical prompt over paraphrases and earlier paraphrases over later ones; we report tie incidence, which is negligible except where noted.

The *self-consistency* condition of size k aggregates the first k canonical samples:

$$\text{SC-}k(q) \;=\; \text{majority}\big(a(M, x_0, 1), \ldots, a(M, x_0, k)\big), \qquad k \le 6. \tag{2}$$

The *paraphrase ensemble* of size k aggregates one decode of each of the first k paraphrases:

$$\text{PE-}k(q) \;=\; \text{majority}\big(a(M, x_1, 1), \ldots, a(M, x_k, 1)\big), \qquad k \le 6. \tag{3}$$

Both conditions spend exactly k model calls and differ only in where the diversity comes from: SC-k perturbs the decoder while the prompt is fixed, whereas PE-k perturbs the prompt while the decoder configuration is fixed. This is the budget match that makes the two conditions comparable.

The *hybrid* condition splits a budget of six calls between the two sources, using two decodes of each of the first three paraphrases:

$$\text{HB-6}(q) \;=\; \text{majority}\big(a(M, x_k, s) : k \in \{1,2,3\},\; s \in \{1,2\}\big). \tag{4}$$

Finally, the *combined* condition aggregates all twelve stochastic votes available in the matrix, $\text{COMB-12}(q) = \text{majority}\big(\text{SC-6}(q) \cup \text{PE-6}(q)\big)$, and serves as a twelve-call reference point for the marginal value of doubling the budget. Three further baselines and controls complete the design. The *single-sample* baseline S1 is one stochastic draw of the canonical prompt at the decoding temperature used throughout ($\tau = 0.7$); its expected accuracy is estimated by averaging over the six canonical samples, and it is the *primary* k=1 baseline of this study, because it is the marginal call that the ensembles actually add to. Its paraphrase-side analogue, PE-1, is the expected accuracy of one paraphrase draw, estimated by averaging over the six first-decode paraphrase calls; the difference PE-1 − S1 measures the *paraphrase difficulty margin*, and the within-family aggregation gains are $\Delta\text{SC} = \text{SC-6} - \text{S1}$ and $\Delta\text{PE} = \text{PE-6} - \text{PE-1}$. All gain statements below state explicitly which of these references they use. The greedy decode $a_g(M, x_0)$ at $\tau = 0$ is reported as a *reference* baseline: it is a different decoding configuration, not a draw from the same distribution, and Section 5.2 quantifies how far it can sit from S1. A *greedy-paraphrase* control re-decodes each paraphrase at $\tau = 0$, giving $\text{PE}^g\text{-6}(q) = \text{majority}\big(a_g(M, x_1), \ldots, a_g(M, x_6)\big)$; comparing PE^g-6 with PE-6 isolates the contribution of prompt-space variation from the decoding stochasticity that the standard paraphrase ensemble mixes in. A chain-of-thought variant repeats the single-call baseline, the self-consistency condition, and (for stem-locus datasets) the paraphrase ensemble under an explicit reasoning instruction, and a hard-task probe repeats the single-call and self-consistency conditions on MMLU-Pro; both are specified in Section 4 and analyzed in Appendices Appendix A–Appendix C.

3.4. Consistency, modal share, and fragility

Three per-question statistics summarize the answer distribution. *Sample agreement* is the frequency of the modal answer among the six canonical

samples,

$$c_{\mathrm{S}}(q) \;=\; \max_{v} \frac{1}{6} \sum_{s=1}^{6} \mathbb{1}\big[a(M, x_0, s) = v\big], \tag{5}$$

and *paraphrase agreement* $c_{\mathrm{P}}(q)$ is defined analogously over the six first-decode paraphrase answers. Both lie in $(0, 1]$, and the value 1 indicates that the corresponding source produced no diversity at all, a situation we call *answer degeneracy*. The *modal vote share* is the corresponding statistic over the union of the twelve stochastic votes, that is, the fraction of all twelve votes that fall on the modal answer, and it is our primary per-question disagreement measure: it reads directly as the strength of the majority (one means unanimous, eleven of twelve means a single dissenting vote, and one half means a tie). We also compute the support-normalized answer entropy of the same twelve-vote distribution, divided by the maximum entropy attainable with the observed number of distinct answers, and report it alongside the modal share for the follow-up conditions, where normalized entropy is the conventional statistic; all entropy values in the paper use this normalization and are reported as fractions, not percentages. The modal share is preferred as the headline statistic because a support-normalized entropy inflates near-unanimous distributions: with twelve votes, a single dissenter already yields a normalized entropy of 0.42, so any entropy threshold below that value merely separates degenerate from non-degenerate questions rather than grading the strength of the majority.

Fragility captures disagreement between the two views of the same question: a question is fragile for a model when the majority answer of the canonical samples differs from the majority answer of the paraphrase answers, that is, when SC-6(q) $\neq$ PE-6(q). Fragility measures how much of the model's answer is an artifact of the particular surface form of the prompt rather than of the underlying question. The greedy-paraphrase control supports a noise-free variant, SC-6(q) $\neq$ PE^{g}-6(q), in which neither side involves stochastic decoding.

3.5. Error detection and selective prediction

We evaluate consistency scores as unsupervised error detectors. For each model, the consistency score of a question (reversed, so that lower agreement indicates higher predicted risk) ranks questions, and we compute the area under the receiver operating characteristic curve (AUROC) for predicting

whether the reference answer is incorrect, where the reference is the combined twelve-vote ensemble unless stated otherwise. When both agreement scores are available, the *combined* detector is their sum of disagreements, $2 - c_{\mathrm{S}} - c_{\mathrm{P}}$, so that a question is flagged when either source shows diversity; this is the synthesis used in the Both columns of Section 5. We complement AUROC with risk-coverage curves that plot the error rate of the accepted subset as the acceptance threshold varies, summarized by the area under the risk-coverage curve (AURC), computed by accepting questions in decreasing order of confidence and averaging the error rate over all prefix sizes; smaller AURC indicates a better ranking, and a random ranking yields AURC equal to the overall error rate. Because several questions typically share the same confidence value, the acceptance order within a tied group is not determined by the score; we therefore average AURC over uniformly random tie-breaking (1,000 permutations), which makes the estimate invariant to the arbitrary ordering of the prediction log. Where a model produces zero errors or constant scores, AUROC is undefined and we report the regime instead; where a model produces fewer than five errors, we report AUROC and AURC descriptively only, without confidence intervals or cross-model comparison.

Because the agreement scores and the reference answer share the same underlying votes, part of the error-detection result is structural rather than empirical: when all twelve votes coincide, the reference equals that unanimous answer by construction, and any detector that reads unanimity is correct by definition on such questions. We therefore separate the degenerate stratum, where the detector outcome is definitional, from the non-degenerate stratum, where the ranking carries empirical content, and we report both. As an additional check against circularity, we recompute the entropy-gain correlation using disjoint votes: entropy over the canonical samples is correlated with the paraphrase-ensemble gain, and entropy over the paraphrase votes with the self-consistency gain, so that the predictor and the outcome share no votes.

3.6. Statistical analysis

Accuracy comparisons between two conditions on the same questions are paired by construction. We report bootstrap 95% confidence intervals for accuracy differences (10,000 resamples, resampling questions) and two-sided exact McNemar tests for paired correctness indicators. Because we interpret a difference of three percentage points or less as practically negligible, we also report two one-sided tests (TOST) against equivalence bounds of ± 3 points,

implemented as bootstrap 90% confidence intervals for the paired difference: equivalence is established only when the entire interval lies inside the bounds. Multiplicity is handled by designation rather than by blanket correction: the four contrasts on the held-out suite that the study is designed to answer are designated *primary* (the pooled gains of SC-6, PE-6, and COMB-12 over S1, and the PE-6 versus SC-6 gap); their two-sided bootstrap p-values are Holm-adjusted within the family and reported alongside the confidence intervals. Every other comparison in the paper — per-model contrasts, per-dataset cells, stratum decompositions, and the entire development suite — is labeled exploratory and carries no correction claim. One caveat is inherent to the design: accuracy moves in units of one question, so the ± 3-point bound corresponds to 2.7 questions at $n = 90$, 2.1 at $n = 71$, and fewer than one question for the smallest completed sets, which makes the TOST verdict on the small models sensitive to the flip of a single question; we flag such cells explicitly. A failure to reject the null hypothesis of no difference is reported as *no detectable difference* and never as evidence of equivalence; at our sample sizes, differences smaller than roughly five points often remain undetectable, and we state this power limitation explicitly wherever it applies. Pooled analyses aggregate (model, question) pairs that share questions across models; because such pairs are not independent, all pooled confidence intervals use a question-level cluster bootstrap that resamples questions and carries every model's pair within a resampled question. Per-model analyses use each model's own completed question set, with the sample size stated in every table. We treat question sets as fixed and do not extrapolate beyond the measured benchmarks.

3.7. When does majority voting help? A heuristic model

The empirical regime structure has a counterpart in a simple plurality-vote model in the spirit of the Condorcet jury theorem [28]. We state it as a heuristic, not a theorem: the asymptotic statements are standard, but the finite-k behavior, the tie handling, and the application to language-model votes each require idealizations that we do not prove. Write p for the probability that one vote drawn from the canonical sampling distribution is the correct answer, and $q_1 \geq q_2 \geq \ldots$ for the probabilities of the incorrect answers, so that $p + \sum_i q_i = 1$. Three idealizations are implicit: the k votes are independent and identically distributed draws from this distribution; plurality ties are broken uniformly at random; and the paraphrase ensemble is described by an analogous distribution of its own, which shares the same gold

answer but need not coincide with the canonical one, because paraphrased prompts are not exchangeable with resampled decodes of the original prompt.

Asymptotic statement. As k grows, the plurality winner converges in probability to the modal answer of the vote distribution. Consequently, when $p > q_1$ the probability that the plurality winner is correct tends to one, and when $q_1 > p$ it tends to zero. The convergence is asymptotic; we do not assert monotonicity in k, and at small k with near-tied leaders the win probability can move in either direction, which is visible in our budget curves as one-question zigzags.

Baseline statement. A single vote is correct with probability p, so majority voting with $k \to \infty$ outperforms *a single stochastic sample* if and only if $p > q_1$, and it underperforms it exactly when the modal answer is wrong. The comparison against greedy decoding is subtler and not covered by the model: greedy decoding approximates the argmax of the model's answer distribution rather than drawing one sample from it, so the greedy answer and the modal sampled answer coincide only when temperature-0 decoding is faithful to the underlying distribution. Section 5.2 shows that the two can disagree by up to ten points on real deployments, which is precisely a violation of this idealization, and it is why we take the single-sample expectation S1, not greedy, as the k=1 baseline.

The model explains the three regimes we observe. In the degenerate regime, p is zero or one and voting is vacuous. In the high-entropy regime with $p > q_1$ (for example Agnes3.0-Flash on logical reasoning), self-consistency converts a distribution that is right on average into a reliable majority. In the high-entropy regime with $q_1 > p$ (GPT-6-Luna and Gemini3.8-Flash on logical reasoning), voting amplifies the modal error and reduces accuracy relative to a single sample, while a second diversity source that shifts the effective vote distribution can restore $p > q_1$, which is what the paraphrase ensemble does for these two models. The model also clarifies why disagreement alone does not predict gains: the modal share measures the spread of the distribution, whereas the sign of $p - q_1$ measures whether the mode is correct, and two distributions with identical shares can sit on opposite sides of that threshold. The budget-matched comparison of Section 5 can then be read as asking which diversity source moves the effective vote distribution further toward the $p > q_1$ region at a given budget.

4. Experimental Setup

4.1. Benchmarks

We assemble a six-benchmark suite that spans knowledge, commonsense, mathematical reasoning, and logical reasoning, with a mix of option counts and answer formats. TruthfulQA-MC1 [29] contributes adversarial factual questions with four or five options; we shuffle option order per question to avoid position bias. PIQA [30] contributes two-choice physical commonsense problems. LogiQA [31] contributes four-choice logical reading comprehension. LSAT logical reasoning and SAT mathematics, both from the AGIEval suite [32], contribute five-choice analytical reasoning and four-choice quantitative items respectively. GSM8K [33] contributes free-form grade-school mathematics problems answered with chain-of-thought reasoning and a final numeric value. We sample fifteen questions per benchmark uniformly at random with a fixed seed, giving ninety questions in total. For the replication study of Section 5.7, a second draw of 119 fresh questions (twenty per benchmark, nineteen for LSAT logical reasoning) is taken from the same raw pools with the same filters by replaying the seeded shuffle and skipping consumed items, so the two sets share a construction procedure and nothing else. The mixture is deliberately diverse in difficulty: as Section 5 shows, the strongest models in our pool saturate five of the six benchmarks while all models find LogiQA and, for some models, GSM8K and SAT mathematics challenging, which supplies both the degenerate and the high-entropy regimes that our analysis requires.

4.2. Models and suites

The study evaluates ten commercial systems from five families, accessed through three providers under their free tiers and organized into two question suites with distinct roles (Section 3). The *development suite* (90 questions, collected October 17–18, 2026) covers six systems: Qwen3.8-Max and Qwen3.8-Omni from the Qwen family [34, 35] on the Aliyun MaaS endpoint; Agnes3.0-Flash and Agnes2.5-Flash from the Agnes family [36] on the Agnes API endpoint; GPT-6-Luna from the GPT family [37, 38] and Gemini3.8-Flash from the Gemini family [39], both on the Apinex endpoint. Its per-model coverage was truncated unevenly by free-tier quotas, to between 12 and 90 complete question matrices. The *held-out suite* (119 fresh questions, collected October 19–23, 2026) re-evaluates the two Agnes models on fresh items and adds four DeepSeek-family systems [40] and Qwen3.7-Flash:

DeepSeekV4.1-Flash, first attempted on Aliyun MaaS (9 questions, retained as a provider record) and completed on the Apinex endpoint; DeepSeekV4-Pro-0813, a pro-tier sibling exposed by the same endpoint; DeepSeekV4-Flash-0731, stopped at 13 questions when its endpoint moved it behind a paid subscription; and Qwen3.7-Flash, stopped at 60 questions when its provider revoked free-tier access. Four systems complete all 119 questions and carry the primary held-out analysis; the two partial systems are reported separately and excluded from tabulation. The models span capability tiers from fast to flagship, including one model that emits an explicit private reasoning trace and one multimodal variant. All systems are queried through OpenAI-compatible endpoints in direct-answer mode on multiple-choice tasks and chain-of-thought on GSM8K; for the two Qwen flagships the proprietary thinking mode is disabled on multiple-choice tasks and enabled on GSM8K. Sampling uses temperature 0.7 with default nucleus probability; greedy calls use temperature 0. Exact endpoint model identifiers, per-call timestamps, and token usage are recorded in the released logs. Because the models are served through third-party endpoints, we cannot verify whether the provider honors the requested temperature for reasoning-style models; the Qwen thinking mode is the one decoding parameter we control explicitly, and findings describe the accessed systems as of October 2026. Table 1 states families, endpoints, access dates, parse rates, and completed coverage per suite.

4.3. Prompt design and paraphrase generation

The canonical prompt for a multiple-choice question consists of the passage when present, the question stem, the option list labeled (A) through (E), and a fixed instruction that requires a direct answer in the form `Answer: <letter>` without explanation. GSM8K prompts instead require step-by-step reasoning ending with `Answer: <number>`. Direct answering on multiple-choice tasks keeps the comparison uniform across models, keeps token costs bounded, and isolates the aggregation question from chain-of-thought length effects; the GSM8K condition covers the reasoning-trace regime.

Paraphrases target the parts of the prompt where semantic-preserving variation is well defined: the question stem for TruthfulQA, LSAT, and LogiQA, and the instruction wording for GSM8K, PIQA, and SAT mathematics, whose stems contain problem-specific content such as numbers that should not be altered. This design yields two paraphrase loci, and Section 5

reports all fragility and gain statistics stratified by locus. Stem paraphrases are generated by Qwen3.7-Flash, prompted to rewrite each stem in six ways while preserving all facts, entities, numbers, and the question type; in the development suite this generator is not part of the evaluation pool. The held-out suite later evaluates Qwen3.7-Flash as a target system (Section 5.7), a dual role we flag here and wherever its rows appear: its paraphrase-condition results carry a self-preference risk, since the model judges its own rewrites, and its row is moreover a partial replication (60 of 119 questions) excluded from the pooled analyses. Instruction paraphrases are six handcrafted variants per dataset that vary wording while keeping the literal answer-format requirement fixed, which keeps answer extraction reliable. As a robustness check, a second generator (Agnes3.0-Flash, outside its own evaluation role) regenerates paraphrases for all stem-locus questions, and the alternative set is validated to the same standard; a planned re-evaluation of the prompt-brittle models under the alternative paraphrases was cut short by provider quota exhaustion and is left to future work.

Every generated paraphrase passes a three-stage validation filter. A lexical stage rejects paraphrases that are identical to the original or whose length ratio falls outside $[0.4, 2.5]$. A semantic stage embeds the original stem and each candidate with a text embedding model and rejects candidates whose cosine similarity to the original falls below 0.75. A form stage, applied to factual questions, rejects rewrites that change the question type, for example from a wh-question to a yes-no question. Candidates that fail any stage are replaced by regenerated candidates until six valid paraphrases per question are obtained; the final set achieves a mean embedding similarity of 0.86 with the originals. One further check strengthens the semantic stage: a bidirectional entailment test, administered by an auxiliary language model, asks whether each paraphrase entails the original and is entailed by it; 267 of 270 final paraphrases pass both directions, and 269 of 270 of the second generator's paraphrases pass, with the full per-item verdicts released alongside the paraphrases. As a sensitivity analysis, the fragility statistics are recomputed under a stricter embedding threshold of 0.80, and the qualitative pattern is unchanged.

4.4. Execution and quality control

The experiment executes the sixteen-call matrix of Section 3 for every question and model through an asynchronous runner with per-provider rate

limiting, automatic retry with exponential backoff, and append-only checkpointing. Answer extraction uses the mandated `Answer:` format first and falls back to scanning the final output lines for a well-formed option letter or number, with the option range constrained by the question. Every stored record includes the extracted answer, its correctness, token usage, and a content snippet for auditing. Extraction succeeds on between 94.7% and 100% of calls across the ten systems (Table 1); cells whose first decode fails to parse are re-drawn under the identical prompt and decoding configuration until extraction succeeds or the free-tier quota of the provider is exhausted, and questions with any remaining unparsed cell are excluded from that model's analysis, so all reported accuracies are computed over complete question matrices. Two safeguards address the residual exclusions. First, a sensitivity analysis recomputes the ensemble conditions under a lenient rule that admits any question with at least four parsed votes per source and aggregates whichever votes parsed; the ordering of the ensemble conditions is preserved for every model, and the denominators grow (for example, from 33 to 42 and 39 questions on the two sources for Agnes3.0-Flash, whose lenient SC-6 and PE-6 reach 88.1% and 92.3% against 90.9% and 90.9% on the strict set); because the added questions were disproportionately ones the provider cut mid-run, they are harder on average, so absolute accuracies on the extended sets sit up to four points below their strict-set values. Second, the excluded questions are compared with the included ones on greedy accuracy to probe for difficulty-related selection bias; no systematic difference in the direction of easier exclusions is observed (Section 5). Per-model completed question counts, which vary between 12 and 90 on the development suite, are stated alongside every result, and pooled analyses use the intersection of completed questions across the models involved, with question-level cluster bootstrap intervals.

4.5. Follow-up conditions

Three follow-up conditions extend the main matrix; all reuse its prompts, paraphrases, and parsing, and all three are analyzed in Appendices Appendix A–Appendix C as descriptive robustness checks rather than primary evidence.

Greedy paraphrases. For every model and every question in its completed set, each of the six paraphrases is decoded once at $\tau = 0$. The control yields the noise-free paraphrase ensemble PE^g-6 and the noise-free fragility

indicator of Section 3, and it covers five of the six models (the free-tier quota of Qwen3.8-Omni was exhausted after completing its main matrix).

Hard-task probe. Because the flagship models saturate most of the main suite, we probe a harder distribution: thirty questions sampled uniformly at random (fixed seed, stratified by subject) from the MMLU-Pro test set [41], which offers ten options per question. The probe administers the single-call greedy baseline and the six-sample self-consistency condition per question, without paraphrases, under the direct-answering instruction. GPT-6-Luna, Agnes2.5-Flash, and Agnes3.0-Flash cover fifteen, thirteen, and eight complete questions respectively; on hard quantitative items the direct-answering instruction does not always suppress reasoning, and outputs truncated at the token cap lost extraction on a majority of Qwen3.8-Max's probe calls, leaving six complete questions (eight with at least four parsed votes) before its quota was exhausted.

Chain-of-thought answering. The main suite answers multiple-choice questions directly, while self-consistency was originally motivated for chain-of-thought reasoning [8, 5]. The follow-up therefore repeats the greedy baseline, the six-sample self-consistency condition, and, for stem-locus questions, the six-paraphrase ensemble under an explicit instruction to reason step by step and end with the mandated answer format. The CoT condition covers the full multiple-choice sets of Agnes3.0-Flash, Agnes2.5-Flash, and GPT-6-Luna, a twelve-question stratified subset for Qwen3.8-Max, and a fifteen-question subset planned for Qwen3.8-Omni whose free-tier quota was exhausted before the condition could run; Gemini3.8-Flash is excluded because its small completed set left no provider capacity after the greedy-paraphrase control. GSM8K is excluded because its main-suite condition is already chain-of-thought.

4.6. Conditions and metrics

Each model and question yields the greedy reference baseline, the single-sample baseline S1, SC-k and PE-k for $k \in \{2, 4, 6\}$, the hybrid HB-6, the combined COMB-12, the agreement statistics c_{S} and c_{P}, the modal vote share, and the fragility indicator. Accuracy is the fraction of questions answered correctly. Error detection is evaluated by AUROC and tie-randomized AURC as defined in Section 3. Budget curves plot accuracy against k for both pure strategies. All analysis code operates on the raw prediction logs, and we release the logs together with the prompts and paraphrases.

5. Results

This section reports the development-suite analyses first (Sections 5.1–5.6); they established the conditions, the regime strata, and the allocation rules, and all of them are exploratory in the sense of Section 3. The held-out evaluation of Section 5.7 then carries the study's primary comparisons on fresh questions. The three follow-up controls (greedy paraphrases, the MMLU-Pro probe, and chain-of-thought answering) are reported in Appendices Appendix A–Appendix C as descriptive robustness checks.

5.1. Overview and data quality

Table 1 summarizes the ten evaluated systems across the two suites; this and the next four subsections report the development suite (six systems, 90 questions, per-model coverage 12 to 90), and Section 5.7 reports the held-out suite. Answer extraction succeeds on between 94.7% and 100% of calls across models; cells whose first decode fails to parse were re-drawn under the identical configuration until extraction succeeded, subject to provider quota. Qwen3.8-Max reached a complete sixteen-call matrix on all ninety questions; Qwen3.8-Omni reached it on seventy-one of the ninety, and the remaining four models on 12 to 40 questions each, bounded by provider rate limits and free-tier quotas. All within-model comparisons are paired and unaffected by coverage, and pooled analyses use the intersection of completed questions, stated explicitly wherever pooled numbers appear.

The sensitivity analysis in Table 1 addresses the residual exclusions from two directions. Under a lenient aggregation rule that admits any question with at least four parsed votes per source, the ordering of conditions is preserved for every model: Agnes3.0-Flash moves from 78.8% greedy to 88.1% under SC-6 and 92.3% under PE-6 on the extended set, and the flagship accuracies shift by at most one question. Comparing excluded with included questions on greedy accuracy shows that the excluded sets are not systematically easier: for Qwen3.8-Omni the excluded questions are nine points *harder* (84.2% versus 93.0%), for Agnes2.5-Flash they are thirty points harder (60.0% versus 90.0%), while for GPT-6-Luna and Gemini3.8-Flash the few excluded questions happened to be answered correctly. The exclusions follow provider rate limits on a fixed question order rather than any adaptive choice, and both checks indicate that the complete-matrix requirement does not manufacture the effects we report.

Table 1: Evaluated systems, endpoints, and coverage across the two suites. Parse rate is the fraction of calls from which the mandated answer format was extracted after retries; n is the number of questions with a complete call matrix. †Also the stem-paraphrase generator; its held-out row carries a self-preference risk (Section 4) and is a partial replication excluded from pooled analyses. ‡Completed on the second endpoint after the first exhausted its free tier. §Partial (13 of 119), excluded from tabulation. All systems run direct-answer on multiple-choice tasks and chain-of-thought on GSM8K.

Panel A: Development suite (Oct 17–18, 2026)

Model	Family / endpoint	Tier	Parse rate	n
Qwen3.8-Max	Qwen / Aliyun	flagship	99.6%	90
Qwen3.8-Omni	Qwen / Aliyun	multimodal	96.8%	71
Agnes3.0-Flash	Agnes / Agnes	fast	94.7%	33
Agnes2.5-Flash	Agnes / Agnes	reasoning	96.9%	40
GPT-6-Luna	GPT / Apinex	reasoning	100.0%	15
Gemini3.8-Flash	Gemini / Apinex	fast	99.6%	12

Panel B: Held-out suite (Oct 19–23, 2026)

Model	Family / endpoint	Tier	Parse rate	n
Agnes3.0-Flash	Agnes / Agnes	fast	98.7%	119
Agnes2.5-Flash	Agnes / Agnes	reasoning	98.1%	119
Qwen3.7-Flash†	Qwen / Aliyun	fast	95.4%	60
DeepSeekV4.1-Flash‡	DeepSeek / Aliyun → Apinex	fast	99.9%	119
DeepSeekV4-Pro-0813	DeepSeek / Apinex	pro	99.9%	119
DeepSeekV4-Flash-0731§	DeepSeek / Aliyun	fast	93.1%	13

5.2. RQ1: Accuracy under budget-matched ensembling

Table 2 reports accuracy for the five aggregation conditions together with the two single-call baselines, and Figure 1 visualizes the four models with at least twenty complete questions. Because the ensembles spend stochastic calls, the single-sample expectation S1 is the primary k=1 baseline, and the Δ columns of Table 2 express both ensemble gains relative to it. Three patterns dominate. First, the two Qwen flagships are at or near ceiling on five of six benchmarks, and for them the pure ensembles reproduce both baselines exactly: SC-6, PE-6, and COMB-12 all return 93.3% for Qwen3.8-Max and 93.0% for Qwen3.8-Omni, identical to greedy and S1. The aggregation strategies are not failing; they are vacuous, because the answer distribution they aggregate contains a single value, a point that Section 5.5 quantifies. Second, measured against S1, no systematic gain for either diversity source is detected in this data: pooled over the four well-covered models (112 model-question pairs on 28 common questions), SC-6 gains +0.9 points and PE-6 loses 0.9 points, with the combined ensemble exactly matching S1; the largest per-model gain over S1 is +1.5 points (Agnes3.0-Flash, both sources). Third, the gains that do exist are concentrated where the modal answer is wrong, and are exploratory at these sample sizes: PE-6 lifts GPT-6-Luna by +10.0 points over S1 (n = 15, CI $[0.0, +22.2]$) and Gemini3.8-Flash by +5.6 (n = 12, CI $[0.0, +13.9]$), while SC-6 *loses* 3.3 and 2.8 points on the same two models (both intervals spanning zero). The wrong-mode condition itself holds clearly only on their LogiQA subsets (five and three questions), so we read the direction as consistent with the plurality-vote signature of Section 3.7 rather than as an established per-model effect.

Table 2: Main results. Accuracy (%) under budget-matched conditions: greedy single call at τ=0, single-sample expectation S1 (mean over the six canonical draws at τ=0.7), six-vote self-consistency (SC-6), six-paraphrase ensemble (PE-6), three-paraphrase two-sample hybrid (HB-6), and the twelve-vote combined ensemble (COMB-12). ΔSC and ΔPE are the gains of SC-6 and PE-6 over the S1 baseline in points.

Model	n	Greedy	S1	SC-6	PE-6	HB-6	COMB-12	ΔSC	ΔPE
Qwen3.8-Max	90	93.3	93.3	93.3	93.3	92.2	93.3	+0.0	+0.0
Qwen3.8-Omni	71	93.0	93.9	93.0	93.0	91.5	93.0	-0.9	-0.9
Agnes3.0-Flash	33	78.8	89.4	90.9	90.9	90.9	90.9	+1.5	+1.5
Agnes2.5-Flash	40	90.0	94.2	95.0	92.5	92.5	95.0	+0.8	-1.7
GPT-6-Luna	15	80.0	76.7	73.3	86.7	86.7	86.7	-3.3	+10.0
Gemini3.8-Flash	12	91.7	86.1	83.3	91.7	91.7	91.7	-2.8	+5.6

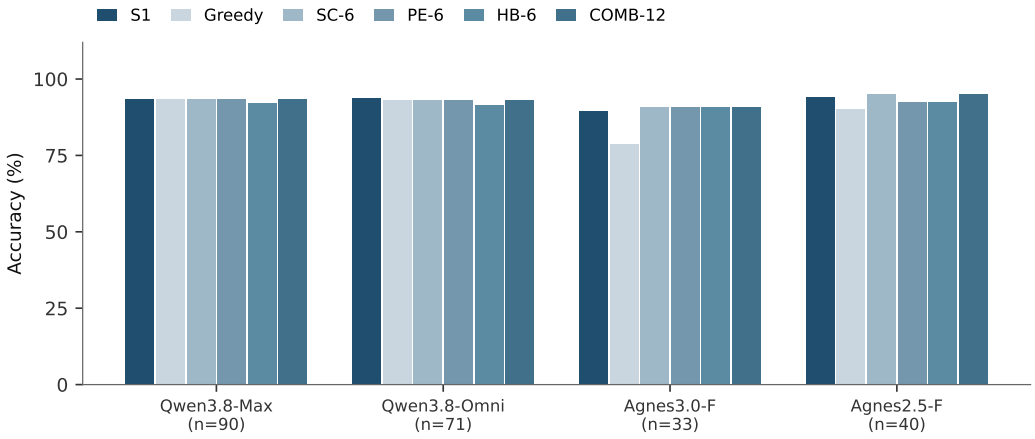


Figure 1: Accuracy of the single-sample baseline S1, the greedy reference, and the four aggregation conditions for the four models with at least twenty complete question matrices. The two Qwen flagships show identical accuracy across conditions, which reflects answer degeneracy rather than agreement between strategies; the models where the modal answer is wrong (GPT-6-Luna, Gemini3.8-Flash) show the two diversity sources pulling in opposite directions. Exact values are given in Table 2.

A separate result concerns the greedy reference itself: greedy decoding at $\tau = 0$ is not an unbiased stand-in for a single stochastic draw. Pooled over the same 112 pairs, S1 reaches 96.4% against 92.0% for greedy, a +4.5 point gap whose question-level cluster-bootstrap interval, $[+0.9, +8.8]$, excludes zero. The gap is model dependent in both direction and mechanism. For the two Agnes models, greedy sits 10.6 and 4.2 points below S1; auditing the discordant questions shows one clear truncation artifact (a greedy chain-of-thought that hit the token cap, after which the parser extracted an intermediate number), two cases of mode locking where the greedy answer differed from every one of the six sampled answers, and remaining cases where greedy picks a minority answer. For GPT-6-Luna and Gemini3.8-Flash the sign reverses (-3.3 and -5.6 points), and for these models the greedy answer always coincided with the modal sampled answer on our data. Neither baseline subsumes the other, which is why Table 2 reports both; but because the ensembles aggregate stochastic calls, S1 is the fair comparison, and gains measured against greedy conflate the ensembling effect with the greedy-versus-sampled gap. For a practitioner who currently deploys only greedy decoding, the most direct consequence is a decoding recommendation rather than an ensembling one: on the affected Agnes endpoints, replacing the greedy call with a single

stochastic draw at the same budget already recovers four to eleven points in our data. The raw outputs of every audited discordant question — the truncation artifact, the mode-locking cases, and the minority picks — are included in the released supplementary material (greedy-bias audit) so that each mechanism can be inspected directly. Section 5.7 adds the decisive qualification: on the held-out suite, pooled greedy and S1 sit within 0.1 points and the gap reappears on only one of four endpoints, so the divergence documented here is a property of specific model–endpoint configurations, not a general property of greedy decoding.

Two further observations complete the accuracy picture. The hybrid allocation is a hedge rather than a dominant strategy: it matches the better pure strategy when one source fails, as for GPT-6-Luna where HB-6 and PE-6 both reach 86.7% while SC-6 falls to 73.3%, but it never exceeds the better pure strategy on any model, and it trails it by exactly one question on Qwen3.8-Max (92.2 versus 93.3), Qwen3.8-Omni (91.5 versus 93.0), and Agnes2.5-Flash (92.5 versus 95.0). The twelve-vote combined ensemble likewise does not dominate the six-vote conditions: pooled it exactly matches S1 at 96.4% against 97.3% for SC-6, because for three of the four pooled models the paraphrase votes it adds are individually no better than a single canonical draw and dilute the majority. Doubling the budget therefore buys no guarantee; the quality of the marginal vote, not the count, governs the outcome.

On the central question, whether one diversity source buys more accuracy per call than the other, no difference is detected in this data against either baseline. Against S1, the pooled SC-6 advantage over PE-6 is 1.8 points — two (model, question) pairs out of 112, both resolved in favor of SC-6 — with a question-level cluster-bootstrap 95% interval of $[0.0, +5.4]$ (the 90% interval that the TOST procedure operates on is $[0.0, +3.6]$), and the per-model intervals for the gain over S1 all include zero (Table 3); no per-model McNemar test on the SC-6-versus-PE-6 discordances approaches significance (all $p \geq 0.5$), and the smallest McNemar p anywhere in the table is 0.125 (SC-6 versus greedy on Agnes3.0-Flash), which does not survive any multiple-comparison correction. We emphasize what this does and does not establish. The TOST procedure at equivalence bounds of ± 3 points does not conclude equivalence for the pooled difference, and the bound itself is coarse at these sample sizes: ± 3 points is 2.7 questions at $n = 90$ and fewer than one question on the smallest sets, so a single flipped question can move the verdict. At these sample sizes the study is powered to detect only large differences; a

genuine advantage of either source of up to several points could go unnoticed. The two strategies are therefore not interchangeable in principle, and for two reasons the data show they are not interchangeable in practice either: they disagree on individual questions at measurable rates (Section 5.3), and their agreement signals catch different errors (Section 5.4).

Table 3: Paired comparisons between conditions: discordant counts (b: first condition right, second wrong; c: the reverse), exact McNemar p, accuracy difference with 95% question-resampled bootstrap CI, and the TOST verdict at ±3 point equivalence bounds. "no" means equivalence is not established at this sample size, not that a difference was found.

Model	n	Comparison	b	c	McNemar p	Diff.	95% CI	TOST ±3
Qwen3.8-Max	90	SC-6 − S1	–	–	–	+0.0	[0.0,0.0]	–
		PE-6 − S1	–	–	–	+0.0	[0.0,0.0]	–
		SC-6 − Greedy	0	0	1.000	+0.0	[0.0,0.0]	–
		SC-6 − PE-6	0	0	1.000	+0.0	[0.0,0.0]	yes
Qwen3.8-Omni	71	SC-6 − S1	–	–	–	-0.9	[-3.3,1.2]	–
		PE-6 − S1	–	–	–	-0.9	[-4.0,1.6]	–
		SC-6 − Greedy	1	1	1.000	+0.0	[-4.2,4.2]	–
		SC-6 − PE-6	1	1	1.000	+0.0	[-4.2,4.2]	yes
Agnes3.0-Flash	33	SC-6 − S1	–	–	–	+1.5	[-2.5,6.1]	–
		PE-6 − S1	–	–	–	+1.5	[-2.5,6.1]	–
		SC-6 − Greedy	4	0	0.125	+12.1	[3.0,24.2]	–
		SC-6 − PE-6	0	0	1.000	+0.0	[0.0,0.0]	yes
Agnes2.5-Flash	40	SC-6 − S1	–	–	–	+0.8	[0.0,2.5]	–
		PE-6 − S1	–	–	–	-1.7	[-5.0,0.0]	–
		SC-6 − Greedy	2	0	0.500	+5.0	[0.0,12.5]	–
		SC-6 − PE-6	1	0	1.000	+2.5	[0.0,7.5]	no
GPT-6-Luna	15	SC-6 − S1	–	–	–	-3.3	[-12.2,4.4]	–
		PE-6 − S1	–	–	–	+10.0	[0.0,22.2]	–
		SC-6 − Greedy	0	1	1.000	-6.7	[-20.0,0.0]	–
		SC-6 − PE-6	0	2	0.500	-13.3	[-33.3,0.0]	no
Gemini3.8-Flash	12	SC-6 − S1	–	–	–	-2.8	[-12.5,4.2]	–
		PE-6 − S1	–	–	–	+5.6	[0.0,13.9]	–
		SC-6 − Greedy	0	1	1.000	-8.3	[-25.0,0.0]	–
		SC-6 − PE-6	0	1	1.000	-8.3	[-25.0,0.0]	no

The comparison is also matched in tokens, not only in calls. Averaged per call, canonical-sample and paraphrase calls cost within one to five percent of each other for every model (for example 223 versus 219 tokens for Qwen3.8-Max and 505 versus 528 for GPT-6-Luna), because validated paraphrases change the surface form but not the length of the task: the mean character-length ratio between paraphrase and original stems is 1.06 for the Qwen

models and 1.13 to 1.14 for the API-served models. Budgets matched in calls are therefore approximately matched in cost for this suite.

5.3. RQ2: Paraphrase fragility

Fragility, the fraction of questions where paraphrasing alone flips the majority answer, is negligible in the degenerate regime and substantial in the high-entropy regime. Qwen3.8-Max shows fragility on 1.1% of questions and Qwen3.8-Omni on 4.2%, concentrated almost entirely in LogiQA, where the respective per-dataset fragility rates are 6.7% and 21.4%. The models with high answer entropy on logical reasoning show more: GPT-6-Luna flips on 13.3% of its questions overall and on 40% of its LogiQA questions, and Gemini3.8-Flash on 8.3% overall and 33.3% of LogiQA. Agnes3.0-Flash, whose gains come with fully consistent votes across both sources, shows zero fragility.

Stratifying by paraphrase locus, which the design fixes in advance, shows that flips are entirely a stem-locus phenomenon. Across all six models, no question ever flips under instruction paraphrasing (zero of 52 pooled instruction-locus pairs), while 3.3% of pooled stem-locus pairs flip, rising to 28.6% and 20% for GPT-6-Luna and Gemini3.8-Flash respectively. The two loci differ by construction in how much of the prompt they vary, so the asymmetry should be read as a property of the perturbation family rather than of the models: instruction rewordings touch only the answer-format scaffolding, while stem rewordings touch the problem content itself. For evaluation practice, the practical consequence is that a single-prompt score on a hard benchmark understates the instability of the measured system by an amount that stem fragility makes visible, and that instruction-level prompt ensembling, as sometimes used to stabilize benchmark scores, probes a much weaker axis of variation than stem-level paraphrasing.

Fragility is not random noise; it tracks the same model-task cells that carry answer entropy and ensembling gains, which supports reading it as an observable symptom of unstable knowledge rather than as a formatting artifact. It is also not an artifact of loose paraphrases: recomputing the statistic with only the paraphrases whose fresh embedding similarity exceeds the stricter 0.80 threshold, requiring at least four such paraphrases per question, leaves the rates essentially unchanged or higher for every model (GPT-6-Luna 33.3% on six qualifying questions versus 13.3% overall; Gemini3.8-Flash 25.0% versus 8.3%; Qwen3.8-Omni 4.8% versus 4.2%), so if anything

the tighter paraphrases flip more. Appendix Appendix A returns to fragility with a decoding-noise-free variant computed from greedy paraphrase decodes.

5.4. RQ3: Consistency as an error signal

Table 4 evaluates agreement scores as unsupervised detectors of ensemble errors, and Figure 2 visualizes AUROC. Models with fewer than five ensemble errors are marked with a dagger and reported descriptively only; at one to three errors, AUROC values are dominated by individual questions and their confidence intervals would span most of the unit interval. The pattern follows the regime structure. For the Qwen flagships, where almost every question is degenerate, sample agreement carries almost no signal (AUROC 0.58 for Qwen3.8-Max, 95% CI $[0.50, 0.78]$) because the score is constant at one for nearly all questions, right or wrong; the degenerate stratum contributes errors that are invisible to any agreement-based detector by construction, and five of Qwen3.8-Max's six errors are missed by both signals. Qwen3.8-Omni retains moderate signal (0.79). In the models with genuine entropy the scores become informative and the stronger source differs across models: for Agnes3.0-Flash, sample agreement is highly discriminative (0.96, CI $[0.89, 1.00]$) while paraphrase agreement is weak (0.60), whereas for GPT-6-Luna paraphrase agreement is the stronger signal (0.92 against 0.58) and for Gemini3.8-Flash both signals separate its single error perfectly. Agnes2.5-Flash, whose answers are mostly degenerate, shows a reversed sample-agreement ranking that we read as noise at its two-error rate.

The risk-coverage analysis in Table 4 quantifies the selective-prediction value of each signal, with ties in the confidence score resolved by uniformly random tie-breaking (Section 3), so each AURC is directly comparable with the model's overall error rate, which is the AURC of a random ranking. Sample agreement attains an AURC of 0.031 for Qwen3.8-Omni against an error rate of 7.0%, and 0.009 for Agnes3.0-Flash against 9.1%; paraphrase agreement attains 0.032 and 0.073 on the same models, so the AURC ordering tracks the AUROC ordering, and each value sits below its model's error rate as expected when the ranking carries signal. For Qwen3.8-Max, whose signals are uninformative, AURC stays at 0.057 against a 6.7% error rate, close to the random baseline as expected. Pooled over the four well-covered models, sample agreement reaches an AUROC of 0.74 (cluster-bootstrap 95% CI $[0.71, 0.75]$) against 0.60 $[0.46, 0.73]$ for paraphrase agreement, an interval

Table 4: Uncertainty estimation. Error rate of the COMB-12 reference; AUROC of reversed agreement scores for predicting ensemble errors (95% bootstrap CI where the model has at least five errors); and the area under the risk-coverage curve (AURC; lower is better), computed by accepting questions in decreasing order of confidence with ties broken uniformly at random and averaged over 1,000 permutations, so a random ranking yields AURC equal to the error-rate column. Both is the sum of the two disagreement scores (AUROC only). †Fewer than five errors: descriptive only. Dashes indicate constant scores or zero errors.

		AUROC			AURC‡	
Model	Err.	c_S	c_P	Both	c_S	c_P
Qwen3.8-Max	6.7	0.583	0.576	0.578	0.057	0.058
Qwen3.8-Omni	7.0	0.788	0.779	0.780	0.031	0.032
Agnes3.0-Flash†	9.1	0.961	0.600	0.922	0.009	0.073
Agnes2.5-Flash†	5.0	0.487	0.704	0.704	0.051	0.029
GPT-6-Luna†	13.3	0.577	0.923	0.769	0.105	0.024
Gemini3.8-Flash†	8.3	0.955	0.955	1.000	0.011	0.011

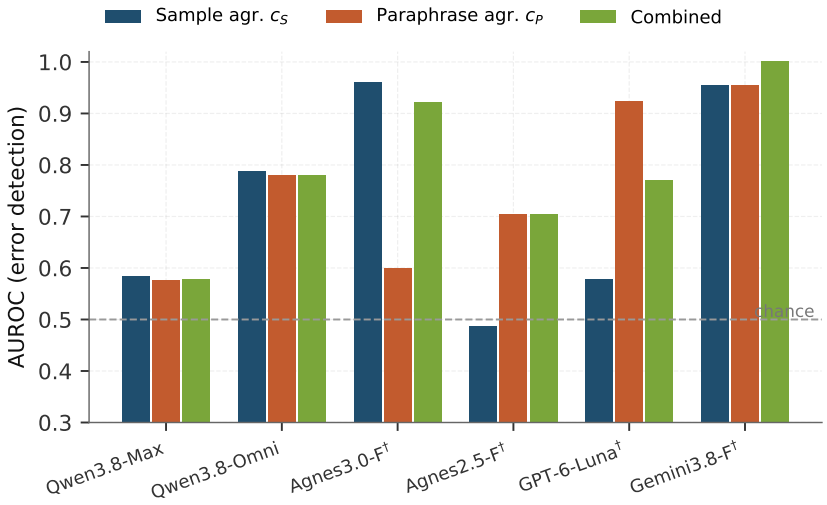


Figure 2: AUROC of reversed agreement scores for detecting ensemble errors. The dashed line marks chance. Signal strength follows the entropy regime, and the stronger source differs across models. †Models with fewer than five errors, reported descriptively.

that overlaps chance: pooled across models with opposite signal strengths, the paraphrase signal averages out.

The two agreement scores therefore measure different failure modes, but the evidence for complementarity is about coverage, not about a superior combined detector. Among the ensemble errors of the models with informative signals, the flagged sets overlap only partially: for Agnes3.0-Flash, sample agreement flags all three errors while paraphrase agreement flags one, two of them only through sample agreement; for GPT-6-Luna, paraphrase agreement flags both errors and sample agreement only one; for Qwen3.8-Omni the two signals flag the same three of five errors. Accordingly, the combined Both score exceeds the best single signal only for Gemini3.8-Flash, ties it for Agnes3.0-Flash (0.92 against 0.96 for sample agreement alone, i.e., slightly below), and falls between the two singles elsewhere; pooled it reaches 0.73 against 0.74 for sample agreement alone. A deployment that can afford both probes obtains partially disjoint evidence, and neither signal subsumes the other, but summing the two disagreement scores is not the way to combine them; a learned or calibrated combination is a natural next step. We also note the structural caveat of Section 3: on questions where all twelve votes coincide, the reference answer equals that unanimous vote by construction, so detectors that read unanimity are correct by definition on the degenerate stratum; the informative content of Table 4 lies in the non-degenerate stratum, which the flagged-error analysis above reflects.

5.5. RQ4: The regime structure

Table 5 reports per model and dataset the greedy accuracy, the single-sample expectation, the combined-ensemble accuracy, the gain of the latter over the former, the mean modal vote share, and the mean dissent share (one minus the modal share), and Figure 3 plots the ensemble gain over S1 against the dissent share for every model-dataset cell. The scatter shows the dependency that anchors the whole study. Disagreement predicts the fragility of the greedy reference rather than the value of ensembling: across the pooled (model, question) pairs of the four well-covered models, the Spearman rank correlation between the dissent share and the gap S1 − greedy is +0.80, meaning that greedy decoding departs furthest from the sampled consensus exactly where the votes disagree, while the correlation between dissent and the COMB-12 gain over S1 is only +0.17: relative to a single stochastic draw, ensembling neither gains nor loses much on average, it only converts an uncertain single draw into the modal answer. Both correlations must

be read with their tie structure in mind: 104 of the 112 pooled pairs are unanimous, on them the gap is nonzero exactly once and the COMB-12 answer never differs from S1, so the rank ordering is determined by the eight non-unanimous pairs, within which the gap correlation is +0.57; using the absolute gap instead of the signed gap leaves the pooled value unchanged at +0.80. Because the combined share and the combined ensemble draw on the same twelve votes, we also recompute the ensemble-gain correlations with strictly disjoint votes: the sample-side dissent share against the PE-6 gain over S1 gives −0.20, and the paraphrase-side dissent share against the SC-6 gain over S1 gives +0.54, both resting on the same eight informative pairs and read as descriptive. The stratified view makes the structure explicit: the pooled pairs split 104 unanimous, five with a strong majority (share at least two thirds), and three with a weak or no majority. In the unanimous stratum the sampled conditions all sit at 98.1% (greedy at 97.1%), and the combined ensemble returns exactly the S1 answer on every one of the 104 unanimous pairs. In the five strong-majority pairs, S1 is 76.7% and the six-vote ensembles 80.0%. In the three weak-majority pairs, S1 is 72.2%, SC-6 reaches 100%, PE-6 falls to 33.3%, and the combined ensemble 66.7%; with three pairs these numbers are descriptive, but they display the full plurality-vote signature of Section 3.7: aggregation pays where the mode is right and is destructive where it is wrong.

Within the high-dissent region, the two largest gains of the study sit on the two cells with the most dissent, and both are carried by the paraphrase side. Under GPT-6-Luna on LogiQA ($n = 5$), the single-sample expectation is 50%, the six-sample majority drops to 40% because the modal sampled answer is wrong, while the paraphrase majority rises to 80% and takes the combined ensemble with it, a +30.0 point gain over S1. The LogiQA cell of Gemini3.8-Flash ($n = 3$) shows the same signature at smaller scale: S1 is 44.4%, the sample majority 33.3%, the paraphrase majority 66.7%, and the combined ensemble follows it for a +22.2 point gain. The plurality-vote model of Section 3.7 summarizes the pattern: disagreement is a necessary condition for aggregation to matter at all, but the sign of the gain is determined by which side holds the correct majority, which the dissent share alone does not measure. Both cells are small — five and three questions — so we read them as the clearest descriptive illustrations of the mechanism rather than as precise estimates of its size.

The budget curves in Figure 4 complete the picture, with the single-sample baseline S1 at $k = 1$ and the greedy level marked as a dotted line;

Table 5: Regime map by model and dataset: greedy accuracy, single-sample expectation S1, COMB-12 accuracy, the COMB-12 gain over S1 (Δ_C), the mean modal vote share over the twelve stochastic votes, and the mean dissent share (one minus the modal share, in percent).

Model	Dataset	n	Greedy	S1	COMB	Δ_C	Share	Diss.%
Qwen3.8-Max	GSM8K	15	100.0	100.0	100.0	+0.0	100.0	0.0
Qwen3.8-Max	SAT-Math	15	100.0	100.0	100.0	+0.0	100.0	0.0
Qwen3.8-Max	PIQA	15	100.0	100.0	100.0	+0.0	98.9	1.1
Qwen3.8-Max	TruthfulQA	15	100.0	100.0	100.0	+0.0	100.0	0.0
Qwen3.8-Max	LSAT-LR	15	100.0	100.0	100.0	+0.0	100.0	0.0
Qwen3.8-Max	LogiQA	15	60.0	60.0	60.0	+0.0	97.2	2.8
Qwen3.8-Omni	GSM8K	15	100.0	100.0	100.0	+0.0	100.0	0.0
Qwen3.8-Omni	SAT-Math	1	100.0	100.0	100.0	+0.0	91.7	8.3
Qwen3.8-Omni	PIQA	11	100.0	100.0	100.0	+0.0	100.0	0.0
Qwen3.8-Omni	TruthfulQA	15	93.3	96.7	93.3	-3.3	96.1	3.9
Qwen3.8-Omni	LSAT-LR	15	100.0	98.9	100.0	+1.1	99.4	0.6
Qwen3.8-Omni	LogiQA	14	71.4	73.8	71.4	-2.4	89.3	10.7
Agnes3.0-Flash	GSM8K	7	85.7	97.6	100.0	+2.4	94.0	6.0
Agnes3.0-Flash	SAT-Math	2	100.0	100.0	100.0	+0.0	95.8	4.2
Agnes3.0-Flash	PIQA	5	100.0	100.0	100.0	+0.0	100.0	0.0
Agnes3.0-Flash	TruthfulQA	5	100.0	100.0	100.0	+0.0	100.0	0.0
Agnes3.0-Flash	LSAT-LR	6	83.3	88.9	83.3	-5.6	95.8	4.2
Agnes3.0-Flash	LogiQA	8	37.5	66.7	75.0	+8.3	84.4	15.6
Agnes2.5-Flash	GSM8K	7	85.7	100.0	100.0	+0.0	98.8	1.2
Agnes2.5-Flash	SAT-Math	11	100.0	100.0	100.0	+0.0	99.2	0.8
Agnes2.5-Flash	PIQA	7	85.7	85.7	85.7	+0.0	98.8	1.2
Agnes2.5-Flash	TruthfulQA	4	100.0	100.0	100.0	+0.0	100.0	0.0
Agnes2.5-Flash	LSAT-LR	6	83.3	83.3	83.3	+0.0	100.0	0.0
Agnes2.5-Flash	LogiQA	5	80.0	93.3	100.0	+6.7	90.0	10.0
GPT-6-Luna	GSM8K	1	100.0	100.0	100.0	+0.0	100.0	0.0
GPT-6-Luna	SAT-Math	5	80.0	80.0	80.0	+0.0	96.7	3.3
GPT-6-Luna	PIQA	2	100.0	100.0	100.0	+0.0	100.0	0.0
GPT-6-Luna	TruthfulQA	2	100.0	100.0	100.0	+0.0	100.0	0.0
GPT-6-Luna	LogiQA	5	60.0	50.0	80.0	+30.0	73.3	26.7
Gemini3.8-Flash	GSM8K	1	100.0	100.0	100.0	+0.0	100.0	0.0
Gemini3.8-Flash	SAT-Math	4	100.0	100.0	100.0	+0.0	93.8	6.2
Gemini3.8-Flash	PIQA	2	100.0	100.0	100.0	+0.0	100.0	0.0
Gemini3.8-Flash	TruthfulQA	2	100.0	100.0	100.0	+0.0	100.0	0.0
Gemini3.8-Flash	LogiQA	3	66.7	44.4	66.7	+22.2	66.7	33.3

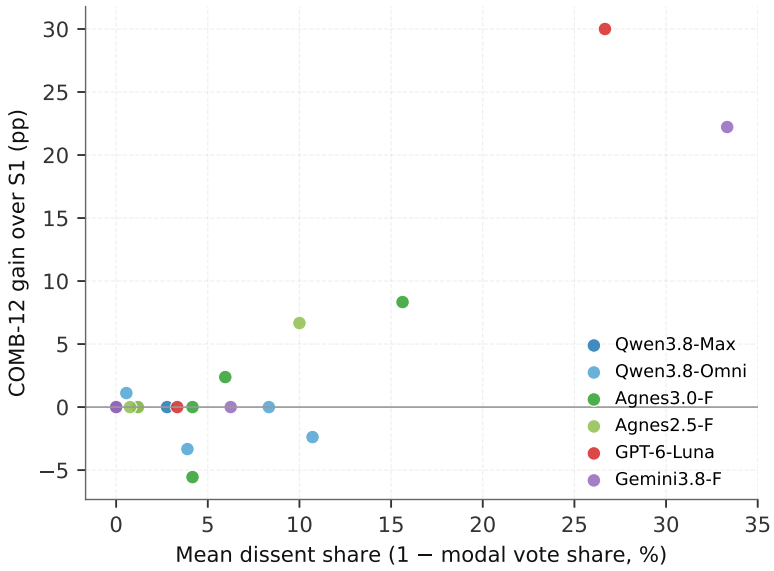


Figure 3: Combined-ensemble gain over the single-sample baseline S1 against the mean dissent share (one minus the modal vote share) for every model-dataset cell. Every zero-dissent cell sits exactly at S1. The largest gains sit at the highest dissent: LogiQA under GPT-6-Luna (+30.0 points, $n = 5$) and LogiQA under Gemini3.8-Flash (+22.2, $n = 3$), where the sample-side modal answer is wrong while the paraphrase side carries the correct majority into the combined vote. Losses appear at moderate dissent (up to −5.6 for LSAT-LR under Agnes3.0-Flash), so the sign of the gain depends on whether the modal answer is correct, not on the amount of disagreement; high-dissent cells contain five questions or fewer, and the pattern is descriptive.

panel headers state how many percentage points one question contributes, since a single flipped question moves accuracy by 1.1 to 3.0 points at these sample sizes. For the flagships the curves are flat at the greedy level within one question at every budget, with the one-question deviations reflecting individual tie-breaks rather than trends: Qwen3.8-Max dips one question under PE at $k \in \{2, 4\}$, and Qwen3.8-Omni wiggles one question above greedy under SC-4. For Agnes3.0-Flash, the paraphrase ensemble reaches its full 90.9% already at $k = 2$ while the sample majority needs $k = 4$ to catch up, the two curves crossing rather than one leading throughout. For Agnes2.5-Flash the ordering is the reverse and persists: sample voting climbs to 95.0% by $k = 6$ while paraphrase voting plateaus at 92.5%. The practical reading is that the choice between the two sources matters less than the choice of whether to aggregate at all, which the disagreement regime determines, and that neither source dominates the other in budget profile across models.

5.6. Budget efficiency: early stopping and adaptive allocation

The regime structure implies that most of a fixed budget is spent confirming answers that additional calls cannot change. Two simulated policies quantify the waste; both are replays of our own data rather than new methods, and both descend from existing work on sample-adaptive stopping: the sequential unanimity rule is a special case of the early-stopping self-consistency line [9, 10], and we read our simulation as a reproduction of that line's finding in the direct-answering, near-degenerate regime rather than as a novel contribution. The unanimity rule draws canonical samples until the first k at which all drawn samples agree, answers with that unanimous vote, and otherwise falls back to the six-sample majority. The adaptive allocation rule of Section 6 probes each question with two canonical samples and two paraphrases, then routes the remaining budget to paraphrases, resampling, or a split according to which side shows diversity. Table 6 reports both policies against the fixed conditions, with the probe's calls charged to the adaptive rule; its mean budget runs 4.1 to 4.7 calls, 68 to 78 percent of the six-call budget.

The unanimity rule preserves SC-6 accuracy within one question for every model at a mean budget of 2.0 to 2.7 calls, because degenerate questions announce themselves after two or three unanimous draws. The adaptive rule matches the best fixed six-call strategy on four of the six models and trails it by exactly one question on the remaining two, Qwen3.8-Omni (91.5 against 93.0) and Agnes2.5-Flash (92.5 against 95.0), while spending a mean of 4.1

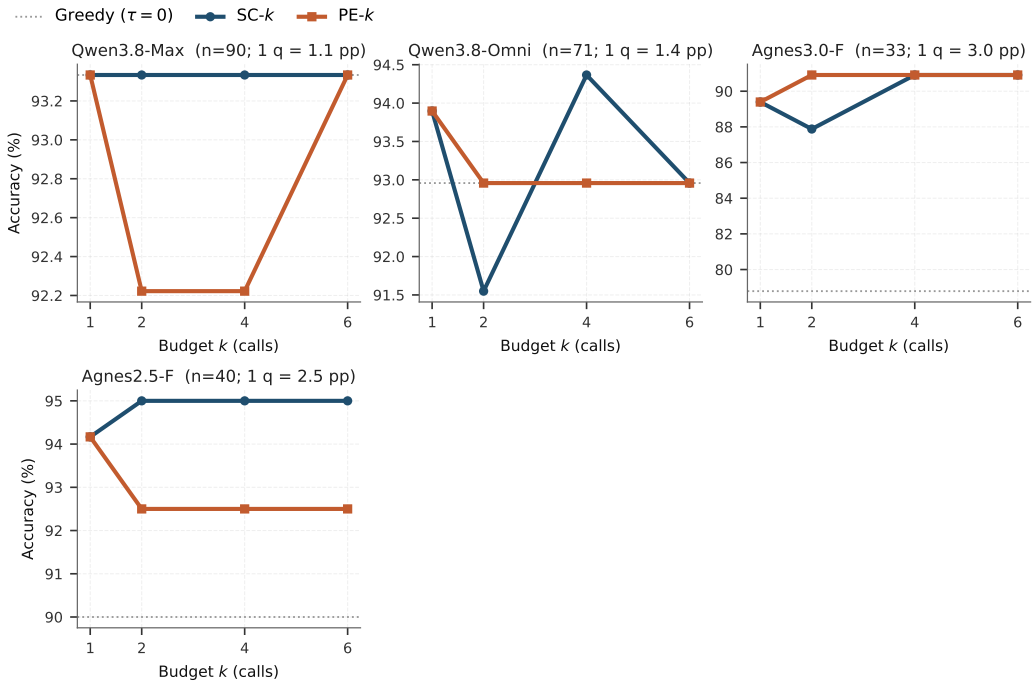


Figure 4: Accuracy against call budget k for self-consistency (SC-k) and paraphrase ensembling (PE-k), with the single-sample baseline S1 at $k = 1$ and the greedy accuracy as a dotted line. Panel headers give each model's question count and the accuracy value of one question. Flat curves indicate the degenerate regime; rising or crossing curves indicate cells where aggregation changes answers.

Table 6: Budget efficiency. Accuracy (%) of the fixed conditions against the sequential unanimity rule (ES, with mean budget $\bar{k}$) and the adaptive allocation rule (Ada, with mean budget and its share of the six-call budget), both simulated on the collected call matrices.

Model	S1	SC-6	PE-6	ES acc	ES $\bar{k}$	Ada acc	Ada $\bar{k}$	Budget	COMB-12
Qwen3.8-Max	93.3	93.3	93.3	93.3	2.0	93.3	4.1	68%	93.3
Qwen3.8-Omni	93.9	93.0	93.0	93.0	2.2	91.5	4.3	71%	93.0
Agnes3.0-Flash	89.4	90.9	90.9	90.9	2.5	90.9	4.6	77%	90.9
Agnes2.5-Flash	94.2	95.0	92.5	95.0	2.1	92.5	4.2	70%	95.0
GPT-6-Luna	76.7	73.3	86.7	80.0	2.0	86.7	4.5	76%	86.7
Gemini3.8-Flash	86.1	83.3	91.7	83.3	2.7	91.7	4.7	78%	91.7

to 4.7 calls, 68 to 78 percent of the six-call budget; it never exceeds the best fixed strategy on any model, and the one-question deficits are exactly the questions whose diversity the two-call probe did not reveal. The comparison against the fixed paraphrase ensemble is the sharpest way to state what the rule adds: the adaptive rule's accuracy equals PE-6's on five of the six models and sits one question below it on Qwen3.8-Omni, so on this data its realized allocation reproduces PE-6 at 68 to 78 percent of its calls. Its increment is the call saving, not accuracy — the accuracy advantage exists only against a misallocated fixed budget such as SC-6 on GPT-6-Luna — and on Agnes2.5-Flash it forgoes the one-question edge that fixed SC-6 holds over PE-6 there. A rule worth its added complexity over always running PE-6 must therefore be justified on fresh questions and harder distributions, where the probe's routing can matter in both directions. Because the rule was designed and evaluated on the same collected matrices, this simulation is a consistency check on the rule, not a validation of it; Section 5.7 performs the fresh-question transfer test. The savings are conservative estimates: the simulation commits to a fixed routing rule after the probe rather than continuing to monitor agreement as later calls arrive, and it charges the full probe even on questions where one call would have sufficed. Both simulations are exact replays of the released call matrices, so they can be re-derived from the logs.

5.7. Held-out evaluation: the fresh-question expansion set

Every headline estimate so far — the unanimity structure, the SC–PE comparison, the adaptive rule — is computed on the same development-suite matrices that motivated the design choices, and per-model coverage there was truncated unevenly by free-tier quota deaths (33 to 90 questions per model). This section therefore carries the study's primary evaluation: a held-out suite that repeats the 16-call matrix on fresh questions, drawn by replaying the seeded shuffle before any of their outputs were inspected (Section 3). It takes the next unseen items from the same raw pools, with the same filters: 20 per benchmark, 19 for LSAT-LR where the filtered pool ran short, identifiers e000–e119 with zero overlap against the development suite. Paraphrases were generated by the same model under the same acceptance filters (embedding cosine at least 0.75, length ratio 0.4–2.5), so the pipeline differs from the main suite only in the identity of the drawn items, and item selection never consulted any model output. Coverage after the provider events of October 2026: Agnes3.0-Flash and Agnes2.5-Flash each completed all 119 questions; DeepSeekV4.1-Flash completed all 119 on a second free endpoint

Table 7: Expansion set (119 fresh questions, 16-call matrix). Accuracy (%) of the fixed conditions, the unanimity rate, and the two rule replays. §: Qwen3.7-Flash completed 60 of 119 questions on a subset that under-represents SAT-Math and GSM8K (Section 5.7); its row is a partial replication, not a full one. Pooled aggregates the four fully covered models over 476 model–question pairs (questions weighted equally, each question contributing the mean over its four models).

		Accuracy (%)							Unanimity rule		Adaptive rule		
Model	n	Greedy	S1	SC-6	PE-6	HB-6	COMB	Unan.%	Acc	$\bar{k}$	Acc	$\bar{k}$	Bud.
Agnes3.0-Flash	119	80.7	79.6	81.5	87.4	83.2	82.4	59	80.7	2.5	81.5	5.2	86%
Agnes2.5-Flash	119	89.9	88.1	90.8	91.6	89.1	89.9	76	90.8	2.3	89.9	4.7	79%
DeepSeekV4.1-Flash	119	73.1	77.6	79.8	81.5	83.2	83.2	58	80.7	2.8	84.0	5.2	87%
DeepSeekV4-Pro-0813	119	79.0	78.0	80.7	82.4	82.4	84.0	61	80.7	2.7	83.2	5.1	85%
Qwen3.7-Flash§	60	85.0	86.1	85.0	86.7	85.0	85.0	82	85.0	2.1	85.0	4.5	74%
Pooled	476	80.7	80.8	83.2	85.7	84.5	84.9	63	83.2	2.6	84.7	5.1	84%

after its original provider exhausted its free tier at 9 questions, so its expansion row is a same-model provider switch rather than a mixed-provider composite (the 9-question original is retained only as a robustness record); DeepSeekV4-Pro-0813, a pro-tier sibling from the same family exposed by the second endpoint, completed all 119 questions there in direct-answer mode after the flash-tier sibling was moved behind a paid subscription mid-collection, so the DeepSeek side of the pool keeps two fully covered models at two capability tiers; Qwen3.7-Flash completed 60 of 119 before its provider revoked free-tier access, and because the runner drains multiple-choice items before GSM8K, the completed subset under-represents SAT-Math (1 of 20) and GSM8K (6 of 20); DeepSeekV4-Flash-0731 stopped at 13 questions when the second endpoint moved it behind a paid subscription mid-collection and is excluded from tabulation. Table 7 reports the tabulated models, the pooled pair statistics over the four fully covered models, and the two rule replays.

On the fresh set, Agnes3.0-Flash shows the first ensemble gain in this study whose confidence interval excludes zero: PE-6 reaches 87.4% against an S1 baseline of 79.6%, a gain of +7.8 points (question-level bootstrap CI +2.4 to +13.7), while SC-6 gains +2.0 points (CI −1.0 to +5.0) and stays statistically indistinguishable from the single-call baseline — exactly the main-suite pattern for sample-space ensembling. COMB-12 (82.4%) lands between its two components, as before. Agnes2.5-Flash, now fully covered, replicates the direction on a higher and flatter baseline: PE-6 91.6% against S1 88.1% (+3.5, CI +0.4 to +7.0), SC-6 90.8% (+2.7, CI +0.7 to +4.9), COMB-12 89.9%. DeepSeekV4.1-Flash contributes the pattern in a different configuration: the pure strategies sit closer to the baseline (PE-6 81.5% against S1

77.6%, +3.9, CI −0.8 to +8.7; SC-6 79.8%, +2.2, CI −0.4 to +4.9), and the two mixed allocations are the conditions whose intervals exclude zero for this model (HB-6 and COMB-12 both 83.2%, +5.6 over S1, CIs +1.5 to +9.8 and +1.7 to +9.7). DeepSeekV4-Pro-0813, the pro-tier sibling completed on the same endpoint, contributes a fourth configuration that sharpens the family pattern: its two individually significant conditions are the combined ensemble and, uniquely in the pool, pure self-consistency (COMB-12 84.0%, +6.0 over S1, CI +2.1 to +10.2; SC-6 80.7%, +2.7, CI +0.1 to +5.5), while the two prompt-space conditions fall between them with intervals spanning zero (PE-6 and HB-6 both 82.4%, +4.3, CIs −0.7 to +9.4 and −0.4 to +9.1). Pooled over the four fully covered models (476 model–question pairs, cluster bootstrap on questions), every six- and twelve-vote condition separates from S1 with an interval excluding zero — SC-6 +2.4 points (CI +1.1 to +3.7), PE-6 +4.9 points (CI +2.3 to +7.6), COMB-12 +4.1 points (CI +2.0 to +6.3) — so on fresh questions both sources of diversity deliver measurable aggregation gains; designated as the study's primary contrasts (Section 3), all three gains survive Holm adjustment within the four-contrast family (adjusted $p = 0.0008$, < 0.0001, and < 0.0001, respectively). The orderings between conditions, however, remain unresolvable at this sample size: PE−SC is +2.5 points (CI −0.6 to +5.7; Holm-adjusted $p = 0.14$), PE−HB +1.3 points (CI −0.4 to +2.9), and COMB−PE −0.8 points (CI −2.9 to +1.3), so no allocation separates from any other; in particular, the mixed allocations are among the best conditions for the two DeepSeek models, which limits how far a pure prompt-space recommendation can be pushed. The Qwen3.7-Flash partial replication behaves like the development-suite flagship models: 82% of its questions are unanimous, every fixed condition lies within 1.1 points of S1, and both rule replays match SC-6 (85.0%) at a fraction of the budget; its paraphrase-condition cells carry the self-preference caveat of Section 4.

A difficulty confound must be separated from the ensembling effect before the margins can be read as aggregation: paraphrases could simply be easier than the original stems, and the comparison against S1 would then mix a difficulty margin with an aggregation margin. Table 8 separates the two, and the pattern holds in all four models. On Agnes3.0 the paraphrases are only marginally easier at the single-call level (PE-1 81.2% against S1 79.6%, +1.7 points), so at most a fifth of its PE-6 margin is paraphrase difficulty, and the within-family aggregation gains are +6.2 points inside prompt space (PE-6 87.4 against PE-1 81.2) against +2.0 inside answer space (SC-6 81.5 against S1 79.6). On Agnes2.5 the difficulty margin is similarly small (PE-1 89.2

against S1 88.1, +1.1), but the two aggregation gains converge (ΔPE +2.4, ΔSC +2.7). DeepSeekV4.1-Flash shows the same configuration at a smaller scale (PE-1 79.1 against S1 77.6, +1.5; ΔPE +2.4, ΔSC +2.2). The pro-tier DeepSeek shifts the balance: its paraphrases are the easiest in the pool at the single-call level (PE-1 81.1 against S1 78.0, +3.1), and its answer-space aggregation margin exceeds its prompt-space margin (ΔSC +2.7 against ΔPE +1.3), so its significant combined condition rests on the two sources jointly rather than on a dominant prompt-space component. Pooled over the four models, six-vote aggregation buys +3.0 points inside prompt space (PE-6 85.7 against PE-1 82.7) against +2.4 inside answer space (SC-6 83.2 against S1 80.8), with the single-call difficulty margin at +1.9 points contributing well under half of the pooled PE-6 advantage; the prompt-space aggregation margin is roughly 1.3 times the answer-space margin, and because +1.9 of the +4.9 headline gap is a difficulty offset, the aggregation-against-aggregation comparison is +3.0 versus +2.4 rather than the raw four-versus-two spread. All margins can be re-derived from the released call matrices.All margins can be re-derived from the released call matrices.

A cross-suite consistency check pools every question completed by the two models covered in both suites (development plus held-out: 152 questions for Agnes3.0-Flash, 159 for Agnes2.5-Flash). The held-out pattern survives the union: on Agnes3.0-Flash, PE-6 sits 6.5 points above S1 (CI +2.2 to +11.1) while SC-6 gains +1.9 (CI −0.7 to +4.4); on Agnes2.5-Flash the two sources converge (+2.2 each, CIs +0.6 to +3.9 and −0.4 to +4.9). The per-model point estimates nonetheless move several points between suites (Agnes2.5-Flash: ΔPE −1.7 on its 40-question development subset, +3.5 on the 119 held-out questions, +2.2 on the 159-question union), a spread that is what unevenly truncated small samples deliver and that cautions against reading any single per-model cell as an effect size; the pooled four-model contrasts, not per-model cells, are where the study's power lies.

The dataset decomposition carries the mechanism, with two cautions stated up front: each cell rests on twenty questions per model (nineteen for LSAT-LR), and all gains below are the within-family margins of Table 8 (ΔSC = SC-6 − S1, ΔPE = PE-6 − PE-1), with single-call difficulty margins reported alongside wherever they are large. On Agnes3.0-Flash's SAT-Math subset, answer-space diversity is close to exhausted: six samples of the original stem move accuracy from 61.7% to 60.0% (ΔSC −1.7), so resampling re-confirms the same wrong computations, while the paraphrase ensemble recovers the subset almost entirely (PE-6 100.0%, ΔPE +17.5); all eight

Table 8: Expansion-set decomposition for the fully covered models: single-call accuracy of original-stem samples (S1) and paraphrase first-samples (PE-1) against the six-vote ensembles (SC-6, PE-6), with the within-family ensembling gains (ΔSC = SC-6 − S1, ΔPE = PE-6 − PE-1), by dataset. Pooled rows weight questions equally.

			Single call (%)		6-vote (%)		Ensembling gain (pp)	
Model	Dataset	n	S1	PE-1	SC-6	PE-6	ΔSC	ΔPE
Agnes3.0-Flash	gsm8k	20	95.0	97.5	95.0	100.0	+0.0	+2.5
Agnes3.0-Flash	logiqa	20	75.0	63.3	90.0	65.0	+15.0	+1.7
Agnes3.0-Flash	lsat_lr	19	86.8	88.6	89.5	94.7	+2.6	+6.1
Agnes3.0-Flash	piqa	20	88.3	86.7	90.0	95.0	+1.7	+8.3
Agnes3.0-Flash	sat_math	20	61.7	82.5	60.0	100.0	-1.7	+17.5
Agnes3.0-Flash	truthfulqa	20	70.8	69.2	65.0	70.0	-5.8	+0.8
	pooled	119	79.6	81.2	81.5	87.4	+2.0	+6.2
Agnes2.5-Flash	gsm8k	20	94.2	95.8	95.0	100.0	+0.8	+4.2
Agnes2.5-Flash	logiqa	20	73.3	77.5	70.0	80.0	-3.3	+2.5
Agnes2.5-Flash	lsat_lr	19	90.4	94.7	100.0	94.7	+9.6	+0.0
Agnes2.5-Flash	piqa	20	94.2	93.3	100.0	95.0	+5.8	+1.7
Agnes2.5-Flash	sat_math	20	98.3	99.2	100.0	100.0	+1.7	+0.8
Agnes2.5-Flash	truthfulqa	20	78.3	75.0	80.0	80.0	+1.7	+5.0
	pooled	119	88.1	89.2	90.8	91.6	+2.7	+2.4
DeepSeekV4.1-Flash	gsm8k	20	98.3	98.3	100.0	100.0	+1.7	+1.7
DeepSeekV4.1-Flash	logiqa	20	52.5	65.0	60.0	65.0	+7.5	+0.0
DeepSeekV4.1-Flash	lsat_lr	19	79.8	84.2	78.9	84.2	-0.9	+0.0
DeepSeekV4.1-Flash	piqa	20	79.2	72.5	80.0	75.0	+0.8	+2.5
DeepSeekV4.1-Flash	sat_math	20	75.8	74.2	75.0	80.0	-0.8	+5.8
DeepSeekV4.1-Flash	truthfulqa	20	80.0	80.8	85.0	85.0	+5.0	+4.2
	pooled	119	77.6	79.1	79.8	81.5	+2.2	+2.4
DeepSeekV4-Pro-0813	gsm8k	20	94.2	95.0	95.0	95.0	+0.8	+0.0
DeepSeekV4-Pro-0813	logiqa	20	54.2	68.3	55.0	70.0	+0.8	+1.7
DeepSeekV4-Pro-0813	lsat_lr	19	78.1	83.3	84.2	84.2	+6.1	+0.9
DeepSeekV4-Pro-0813	piqa	20	90.0	83.3	95.0	80.0	+5.0	-3.3
DeepSeekV4-Pro-0813	sat_math	20	69.2	79.2	70.0	85.0	+0.8	+5.8
DeepSeekV4-Pro-0813	truthfulqa	20	82.5	77.5	85.0	80.0	+2.5	+2.5
	pooled	119	78.0	81.1	80.7	82.4	+2.7	+1.3
Pooled	gsm8k	80	95.4	96.7	96.2	98.8	+0.8	+2.1
Pooled	logiqa	80	63.8	68.5	68.8	70.0	+5.0	+1.5
Pooled	lsat_lr	76	83.8	87.7	88.2	89.5	+4.4	+1.8
Pooled	piqa	80	87.9	84.0	91.2	86.2	+3.3	+2.3
Pooled	sat_math	80	76.2	83.8	76.2	91.2	+0.0	+7.5
Pooled	truthfulqa	80	77.9	75.6	78.8	78.8	+0.8	+3.1
Pooled	pooled	476	80.8	82.7	83.2	85.7	+2.4	+3.0

SAT-Math questions on which SC-6 and PE-6 disagree are won by the paraphrase vote. Because SAT-Math paraphrases vary the instruction wording rather than the stem, part of this cell is plausibly an instruction effect rather than an aggregation effect: the paraphrase single call already sits 20.8 points above S1 there (PE-1 82.5 against 61.7), which suggests that the canonical directive to answer without explanation is itself suboptimal for quantitative items and that some instruction variants elicit more careful behaviour. The within-family margin is net of this base-rate shift, but separating instruction quality from aggregation proper would require re-running self-consistency under the best instruction variant, which we leave to future work. GSM8K shows the same direction in weaker form (PE-6 100.0%, ΔPE +2.5 on a floor of 95.0%). LogiQA is the mirror image for this model: paraphrases are 11.7 points harder at the single-call level (PE-1 63.3 against S1 75.0), indicating that restating a verbal-reasoning stem can shift the reasoning problem itself, and there the sample vote is the only condition that aggregates a gain (SC-6 90.0%, ΔSC +15.0 against ΔPE +1.7; five of the seven flips favour SC). TruthfulQA is the one dataset where its six-sample vote actively hurts (SC-6 65.0 against S1 70.8), consistent with the development-suite reading that on knowledge questions the modal answer can be a confident misconception. Agnes2.5-Flash contributes the counter-case that bounds the mechanism: its SAT-Math and GSM8K floors sit near ceiling (S1 98.3 and 94.2), its pooled prompt-space aggregation margin is correspondingly small (ΔPE +2.4), and on LogiQA the Agnes3.0 ordering reverses (PE-6 80.0 against SC-6 70.0, ΔPE +2.5 against ΔSC −3.3), so on stem-locus datasets the PE–SC ordering is a property of the model–paraphrase interaction rather than of the dataset. The DeepSeek models sharpen the same caution with opposite difficulty signs: DeepSeekV4.1's LogiQA paraphrases are 12.5 points easier at the single-call level (PE-1 65.0 against S1 52.5) — where Agnes3.0's were 11.7 points harder — and there the sample vote is again the only condition that aggregates a gain (ΔSC +7.5 against ΔPE +0.0), while its SAT-Math repeats the computation-deterministic direction (ΔSC −0.8, ΔPE +5.8) and its TruthfulQA supplies the counter-case to Agnes3.0's, with the sample vote helping (ΔSC +5.0). The pro-tier DeepSeek shows the pool's largest single-call difficulty margins in both directions. Its LogiQA paraphrases are 14.1 points easier (PE-1 68.3 against S1 54.2); the paraphrase ensemble holds that base-rate advantage through voting (PE-6 70.0 against SC-6 55.0), but the within-family gains are small (ΔPE +1.7 against ΔSC +0.8), so what moves this cell is the difficulty margin rather than aggregation — the mirror

image of Agnes3.0's LogiQA. On PIQA its paraphrases are 6.7 points harder and the paraphrase ensemble loses to the sample ensemble by fifteen points (PE-6 80.0 against SC-6 95.0; measured against S1 the paraphrase side loses 10.0 points, of which 6.7 is the harder single-call base and −3.3 the within-family change, while the sample side gains ΔSC +5.0). Pooled over the four models, the computation-deterministic datasets carry the paraphrase aggregation advantage (SAT-Math ΔPE +7.5 against ΔSC +0.0; GSM8K +2.1 against +0.8), while on LogiQA the sample side aggregates the larger gain (ΔSC +5.0 against ΔPE +1.5), the paraphrases' pooled single-call advantage there (+4.7 over S1) doing more of the work than aggregation, and TruthfulQA is neutral in aggregate (ΔSC +0.8, ΔPE +3.1). Two readings of these cells deserve explicit caution. First, the single-call difficulty margins of the same dataset's paraphrases swing by more than twelve points across models (LogiQA: −11.7 on Agnes3.0, +12.5 on DeepSeekV4.1, +14.1 on the pro tier) — too large to dismiss as sampling noise at these denominators, and consistent either with specific rewrites shifting the task for specific models or with equivalence failures that the aggregate validation filters cannot see; the model–paraphrase interaction is real but not currently predictable, and we treat it as descriptive. Second, because each cell is a twenty-question stratum, individual margins carry wide intervals; the transferable content is the pair of directions — paraphrase aggregation pays where resampling is exhausted and reverses where paraphrases shift the task — rather than any single cell's magnitude.

The strata defined on the development suite also organize the held-out votes, and the decomposition — computed on the same twelve votes that define the pooled contrasts — locates where the (statistically unresolved) PE−SC margin concentrates. The weak-majority stratum (mode share below two-thirds) holds 24, 8, 14, and 15 questions for the four models: PE-6 scores 58.3% against 33.3% for SC-6 and 31.9% for S1 on Agnes3.0, 50.0% against 25.0% and 27.1% on Agnes2.5, 50.0% against 35.7% and 33.3% on DeepSeekV4.1, and 53.3% against 33.3% and 31.1% on the pro tier; averaging the four models' strata (61 pairs in total) gives PE-6 52.9% against 31.8% for SC-6 and 30.9% for S1, a +22-point margin exactly where answer-space votes have the least to aggregate, and for the two DeepSeek models the combined twelve-vote ensemble adds a further margin on top (COMB-12 57.1% against PE-6 50.0% on V4.1; 60.0% against 53.3% on the pro tier). We emphasize what this decomposition is and is not: it reuses the same votes that produce the pooled PE−SC gap of +2.5 points (interval touching zero), so

it is a localization of that gap, not an independent replication of it, and part of the concentration is mechanical — a question on which the sample side is degenerate but wrong is exactly a question where only the paraphrase side can move the answer, so the stratum where PE can win is selected, in part, by the failure of SC. The unanimous stratum ($n = 70$, $n = 90$, $n = 69$, and $n = 73$) is saturated at 96.3% pooled under every condition, and the strong-majority stratum ($n = 25$, $n = 21$, $n = 36$, and $n = 31$) sits between, with the two six-vote ensembles at 76–77% against an S1 of 67.0%. The $\tau = 0$ anomaly, by contrast, appears on exactly one of the four models, and the split is informative. On the two Agnes endpoints S1 minus greedy is -1.1 and -1.8 points, and on the pro-tier DeepSeek -1.0. On the direct-answer DeepSeekV4.1 endpoint the development-suite signature reappears: greedy sits 4.5 points below S1 (73.1% against 77.6%), and auditing its 16 off-mode greedy answers shows 13 of them wrong, with completion lengths of one to eleven characters — genuine divergence between the temperature-zero answer and the sampled mode, not the truncation artifact of the development suite; the pro-tier sibling's 14 off-mode answers split 10 wrong against 4 right, a third clean case. Pooled, 50 of 476 greedy answers (10.5%) fall outside the six-sample mode. This relocates the greedy audit's lesson: the temperature-zero failure is a property of the model–endpoint configuration rather than of the question draw or of a universal decoding pathology — absent on three of four held-out endpoints, present without truncation on the fourth — so practitioners should verify greedy against the sampled expectation per deployment rather than assume either baseline.

The adaptive allocation rule was designed on the development-suite matrices, so the held-out set supplies the fresh-question transfer test that Section 5.6 called for. On three of the four fully covered models the rule transfers in budget but not in allocation. It reproduces SC-6 exactly (no question gap) at 86% of the six-call budget on Agnes3.0 (81.5% at a mean of 5.2 calls), and it matches SC-6 exactly at 74% of budget on the Qwen partial (85.0%). But it does not capture the prompt-space margin there: it sits seven questions below PE-6 on Agnes3.0, because questions whose two-call probe shows diversity on both sides are routed to the hybrid vote, which underperforms a pure paraphrase vote on this distribution. On Agnes2.5 the rule lands one question below SC-6 (89.9% against 90.8%) and two below PE-6, at 79% of budget (a mean of 4.7 calls); its probe routes 97 of 119 questions to the four-call unanimous exit, 10 to the sample vote, 9 to the paraphrase vote, and 3 to the hybrid. On DeepSeekV4.1 the rule does what no fixed condi-

tion does there: 84.0% at a mean of 5.2 calls (87% of budget), five questions above SC-6, three above PE-6, and one above HB-6 and COMB-12, the only model in either suite on which the rule exceeds every fixed allocation — though the margin over the best of them is a single question, so we read it as reaching the fixed frontier rather than dominating it; its probe routes 83 of 119 questions to the unanimous exit, 18 to the paraphrase vote, 12 to the sample vote, and 6 to the hybrid. On the pro-tier DeepSeek the rule lands at 83.2% at a mean of 5.1 calls (85% of budget), three questions above SC-6 and one above PE-6 but one question below its best fixed condition, the combined ensemble (84.0%); its probe routes 86 of 119 questions to the unanimous exit, 20 to the paraphrase vote, 9 to the sample vote, and 4 to the hybrid. Pooled over the four models the rule returns 84.7% at a mean of 5.1 calls (84% of budget). The development-suite conclusion therefore needs its held-out amendment: the rule saves 13 to 21 percent of the calls, exceeds every fixed allocation on one model (by a single question), loses one question to its best fixed condition on the pro tier and two to PE-6 on Agnes2.5, and sits seven questions below the paraphrase vote on Agnes3.0 — routing toward paraphrases when answer-space diversity is low remains the regime where the most is gained, and it is exactly the regime the probe's two-call answer-space check undersamples.

These results qualify, without overturning, the development-suite null. Read together, the two suites say that six-call sample-space ensembling moves accuracy little for any model in either suite (per-model ΔSC between -3.3 and $+2.7$ points), while prompt-space ensembling produces the largest well-powered gains — pooled $+4.9$ points over S1 on the held-out set (CI $+2.3$ to $+7.6$), of which $+1.9$ is a single-call difficulty offset and $+3.0$ the within-family aggregation gain, against $+2.4$ for self-consistency (CI $+1.1$ to $+3.7$) — and it does so only where two conditions hold jointly: answer-space diversity is exhausted, as on computation-deterministic tasks where six samples rederive the same wrong arithmetic, and paraphrases preserve task semantics. Where paraphrases distort the task, as on Agnes3.0's LogiQA subset or the pro-tier DeepSeek's PIQA subset, the advantage reverses. Which allocation realizes the margin is model-dependent — the pure paraphrase vote on the two Agnes models, the hybrid and combined allocations on DeepSeekV4.1, the combined ensemble and self-consistency on its pro-tier sibling — so the transferable content is the two qualifying conditions rather than a recipe. The gain is established here for four fully covered models from two providers and two families (two Agnes generations, two DeepSeek tiers) on one fresh

draw, so we read it as a demonstrated possibility with an identified mechanism rather than as a universal effect.

6. Discussion

6.1. Principal findings

The study yields four principal findings. First, the effectiveness of every test-time aggregation strategy is governed by a precondition that the literature rarely states explicitly: the model must produce diverse answers at all. For the flagship models in our pool, five of six benchmarks sit in a degenerate regime in which six stochastic samples and six paraphrases return the same answer for almost every question, and in that regime self-consistency, paraphrase ensembling, and the combined ensemble reproduce both single-call baselines exactly, with the hybrid allocation differing by at most one question. This part of the degeneracy result is structural rather than empirical: when all votes coincide, every aggregation of them returns the same answer by construction. The empirical content lies in how common the coincidence is, and in its consequence that agreement-based uncertainty scores are uninformative exactly there. Second, measured against the correct k=1 baseline, the single-sample expectation S1, the two diversity sources deliver small, reliable, and statistically inseparable aggregation gains on the held-out suite: pooled over its four fully covered models (476 model–question pairs), self-consistency gains +2.4 points over S1, the paraphrase ensemble +4.9, and the twelve-vote combination +4.1, all with intervals excluding zero and surviving Holm correction within the designated primary family (Section 3); the paraphrase margin over self-consistency (+2.5, CI −0.6 to +5.7) does not separate from zero, and after removing the paraphrase difficulty margin (+1.9 at the single-call level) the like-for-like aggregation comparison is +3.0 against +2.4 — roughly a factor of 1.25, not the factor of two that the raw headline suggests. On the development suite, whose per-model coverage was truncated unevenly (12 to 90 questions), the same pooled comparison sits at noise (SC-6 +0.9, PE-6 −0.9 over 112 pairs), and the per-model gains measured against greedy decoding are largely artifacts of a greedy reference that diverges from the sampled expectation on specific endpoints; we therefore treat the held-out suite as the primary evidence and the development suite as the design phase whose role was to fix the conditions, strata, and rules. Where the modal answer is wrong, the two sources pull in opposite directions, and the sign of the margin is model- and dataset-conditional: the paraphrase

vote wins where resampling is exhausted, as on computation-deterministic tasks, and reverses where paraphrases shift verbal semantics (LogiQA for one model, PIQA for another), and the allocation that realizes the gain varies across models. Third, paraphrase agreement functions as an error signal in its own right, and its error coverage overlaps only partially with that of sample agreement: across models, the stronger source differs, so neither signal subsumes the other, and a deployment that can afford both probes obtains partially disjoint evidence, although the simple sum of the two disagreement scores does not dominate the best single signal in our data. Fourth, fragility, the fraction of questions whose answer flips under paraphrasing alone, tracks answer disagreement, concentrates entirely in stem-locus paraphrasing, and appears in the same model-task cells that carry disagreement, which supports the interpretation that paraphrase disagreement and sampling disagreement are two observable symptoms of the same underlying instability.

6.2. A practical allocation rule, and how it performs

The regime structure suggests a simple procedure for practitioners with a fixed per-question budget. Probe the question first with a small number of calls, compute sample agreement and paraphrase agreement, and branch on the result. If both agreement scores equal one, additional calls of either kind are unlikely to change the answer, and the budget is better spent elsewhere; the degenerate regime is identifiable from a handful of calls at negligible cost. If sample agreement is high but paraphrase agreement is low, the model is prompt-brittle, and the remaining budget should go to paraphrases. If the reverse holds, the budget should go to resampling. If both scores are low, split the remaining budget between the two sources.

We evaluated this rule by simulation on the collected call matrices, charging the probe's four calls (two canonical samples and two paraphrases) to the adaptive condition's budget. The adaptive rule matches the best fixed six-call strategy on four of the six models and trails it by exactly one question on the other two (Qwen3.8-Omni and Agnes2.5-Flash), while spending a mean of 4.1 to 4.7 calls per question, 68 to 78 percent of the six-call budget; where it wins, it wins against a misallocated fixed budget, as for GPT-6-Luna, where it returns 86.7% at a mean cost of 4.5 calls against 73.3% for fixed SC-6. Stated against the fixed paraphrase ensemble rather than the best fixed strategy, the rule's realized allocation equals PE-6's accuracy on five of the six models at 68 to 78 percent of its calls, sitting one question below it on Qwen3.8-Omni, so its measured increment over simply running PE-6 is the

call saving rather than accuracy; the hedge it buys is robustness against the opposite allocation error, at the price of forgoing SC-6's one-question edge on Agnes2.5-Flash. Because the rule was designed and evaluated on the same collected matrices, this simulation is a consistency check, not a validation; the held-out set of Section 5.7 supplies the transfer test, and there the rule's behaviour is model-conditional rather than uniformly conservative: on three of the four fully covered models it transfers in budget but not in allocation — it reproduces SC-6 exactly at 86 percent of the six-call budget on Agnes3.0-Flash while sitting seven questions below its PE-6, because the regime that rewards paraphrase allocation is exactly the one its two-call probe routes to the hybrid vote; it trails PE-6 by two questions on Agnes2.5-Flash at 79 percent; and on the pro-tier DeepSeek it sits one question below the combined ensemble at 85 percent — while on DeepSeekV4.1-Flash it exceeds every fixed allocation (84.0 percent at 87 percent of budget, one question above the best of them), the only instance in either suite where the probe converts into a win over the whole fixed frontier. Across the development suite's six models, an even simpler sequential rule that draws canonical samples until the first unanimity and otherwise falls back to the six-sample majority preserves SC-6 accuracy within one question everywhere at a mean budget of two to three calls; this is a special case of the early-stopping self-consistency family [9, 10], and our simulation reproduces that family's finding in the direct-answering, near-degenerate regime. Because agreement statistics are cheap to compute online, the rules add no model training and no human labeling, and both simulations are reproducible from the released logs. The hybrid allocation, by contrast, is best understood as the fixed-budget version of the same hedge: it matches the best pure strategy when one source fails and trails it by about one question when both work, so the adaptive probe matches it in our data at two-thirds to three-quarters of the cost.

6.3. Implications for evaluation and for uncertainty estimation

Two methodological consequences follow. For benchmark evaluation, our results add an aggregation dimension to the multi-prompt argument of Mizrahi et al. [1]: the same question answered under one prompt with one decode, under six paraphrases, or under twelve combined calls yields systematically different accuracies in the high-entropy regime, so reported scores should state the aggregation configuration, and comparisons across papers that aggregate differently are not like for like. The single-sample baseline S1 deserves the same reporting discipline, because greedy decoding and a single

stochastic draw differ by four to eleven points on three of the ten model–endpoint configurations we measured, in both directions, and are statistically indistinguishable on the rest, including three of the four held-out endpoints; the divergence is an endpoint property to be checked per deployment, not a universal offset. For uncertainty estimation, the degenerate regime is a warning whose scope we state precisely. Our measurements cover multiple-choice questions answered directly, where sampling-based confidence measures, including simple agreement heuristics, are uninformative when the sample distribution collapses, and our data show that this collapse is the common case for strong models on saturated tasks. Semantic-entropy methods for free-form generation [6, 7] face an analogous risk when the sampled paraphrase classes collapse, but our design does not test that setting, and the caution transfers by argument rather than by measurement. Paraphrase probes partially restore the signal because they perturb the input rather than the decoder, but they too inherit the collapse when the model is robust in both views, as our flagship models are on five of six benchmarks. Confidence methods should therefore be validated per regime rather than averaged over mixed test sets, and deployments should verify that the answer distribution carries variance before trusting agreement-based scores.

6.4. Limitations

Eight limitations bound the scope of our conclusions. First, sample size and power: fifteen questions per benchmark and completed sets of 12 to 90 questions per model support paired within-model comparisons of large effects, but the cluster-bootstrap intervals are wide, TOST does not establish equivalence for most pairs, and differences of a few points between strategies would go undetected; the 119-question held-out set of Section 5.7 multiplies per-model coverage for four systems (two Agnes generations and two DeepSeek tiers), while two of its six systems remain partially covered, and scaling each benchmark to hundreds of questions remains the single change most likely to sharpen the conclusions. Second, provider rate limits and free-tier quotas constrained coverage unevenly across models; the quota of Qwen3.8-Omni was exhausted before its follow-up conditions could be run, and during the expansion collection the shared workspace serving Qwen3.7-Flash and both DeepSeek models revoked free-tier access mid-run: Qwen3.7-Flash stopped at 60 of 119 questions, DeepSeekV4.1-Flash was completed on a second free endpoint after the first died at 9 questions, and DeepSeekV4-Flash-0731 stopped at 13 when the second endpoint moved it behind a paid

subscription mid-collection, an instance of the broader reproducibility hazard that free-tier access to a given model–endpoint configuration can be withdrawn at any time; coverage follows a fixed question order rather than adaptive selection, but cross-model aggregates weight models unequally. Third, the models are commercial systems accessed through third-party endpoints: we cannot verify that the requested temperature is honored, especially for reasoning-style models, and decoding internals, post-training, and version pinning are outside our control; the greedy-versus-S1 gap documented in Section 5.2, including its truncation artifact, is itself a manifestation of this limitation, and findings describe the accessed systems as of October 2026 and need not transfer to other versions or serving configurations, although we release per-call timestamps and token usage so the configuration is auditable. Fourth, paraphrases were generated by two models and validated by embedding similarity with a fixed threshold plus a bidirectional entailment check, but semantic equivalence of adversarial questions is not fully formalizable, and stricter thresholds or human-curated paraphrases could shift the fragility statistics; a related caveat is that Qwen3.7-Flash, the stem-paraphrase generator, is itself evaluated in the held-out suite, so its paraphrase-condition cells carry a self-preference risk, and its partial coverage (60 of 119) keeps them out of the pooled analyses. Fifth, the multiple-choice tasks use direct answering while GSM8K uses chain of thought; the aggregation comparison is internally valid within each answering mode, and the follow-up condition of Appendix Appendix C measures the mode effect directly, but absolute accuracies are not comparable across modes. Sixth, the high-dissent regime is sampled through fewer model-task cells than the degenerate regime, resting primarily on logical reasoning and selected quantitative tasks; the MMLU-Pro probe of Appendix Appendix B begins to broaden this, at small scale. Seventh, error-detection metrics are computed against ensemble references that share votes with the detectors, and the adaptive allocation rule was designed and evaluated on the same development-suite matrices, so both carry a same-data caveat; the held-out set mitigates but does not eliminate it, because the strata and the rule were fixed after the development-suite results were known; we separate the definitional degenerate stratum and report disjoint-vote correlations, and the held-out set of Section 5.7 now provides the fresh-question evaluation of the allocation rule, but an error reference from fully independent draws would still be cleaner. Eighth, all conditions use a fixed budget of six or twelve calls with majority voting; ranked voting,

weighted aggregation, or learned vote combination could change the budget-matched comparison.

6.5. Future work

Three extensions follow naturally. The adaptive allocation rule can be moved online, interleaving probing and allocation per question rather than simulating the split afterwards, and its threshold can be tuned against a target budget rather than fixed at unanimity. The comparison can be extended to free-form generation, where semantic clustering replaces majority voting and paraphrase ensembles interact with surface-form competition effects [42], and where the degeneracy phenomenon has direct consequences for semantic-entropy hallucination detection that our multiple-choice design cannot measure. Finally, the fragility statistic can serve as a benchmark diagnostic in its own right, complementing accuracy with a measure of how strongly a model's answers depend on the surface form of the question.

7. Conclusion

We presented a budget-matched comparison of the two diversity sources available to test-time ensembling for large language models: resampling the decoder on a fixed prompt, and perturbing the prompt across validated semantic paraphrases. A sixteen-call matrix per question, applied to ten commercial systems from five families on six benchmarks, allowed every condition — self-consistency, paraphrase ensembling, a hybrid allocation, and a combined ensemble — to be derived from matched calls on identical questions, with greedy-paraphrase, chain-of-thought, and hard-task controls reported in the appendices. The study proceeded in two phases. A development suite of 90 questions, covered unevenly (12 to 90 questions per model) by free-tier quota events, fixed the conditions, the regime strata, and the allocation rules; on it, no systematic gain for either source over the single-sample expectation S1 was detectable. The primary evidence is a held-out suite of 119 fresh questions fully covered by four models from two providers and two families. There, both sources help: pooled, self-consistency gains +2.4 points over S1 (CI +1.1 to +3.7), the paraphrase ensemble +4.9 (CI +2.3 to +7.6), and the twelve-vote combination +4.1 (CI +2.0 to +6.3), all surviving Holm correction; the two sources remain statistically inseparable from each other (+2.5, CI −0.6 to +5.7), and +1.9 of the paraphrase headline is a single-call difficulty offset, leaving +3.0 of within-family aggregation against +2.4

for resampling. The margin is conditional in a way that the title question's answer must reflect: prompt-space ensembling pays where answer-space diversity is exhausted — on computation-deterministic tasks, where six samples re-derive the same wrong arithmetic and paraphrases recover it — and reverses where paraphrases shift the task's verbal semantics; the allocation that realizes the gain is model-dependent, with the pure paraphrase vote best on the two Agnes models and mixed or combined allocations best on the two DeepSeek tiers; and in the degenerate regime that dominates current benchmarks for confident models, neither source pays, agreement-based uncertainty estimates lose their signal, and the budget is better saved. Greedy divergence from the sampled expectation proved to be an endpoint property — four to eleven points on three of ten model–endpoint configurations, within noise elsewhere — so we recommend reporting the single-sample expectation alongside every greedy score and verifying the pair per deployment. An adaptive probe-and-route rule, simulated on the collected matrices, matched the fixed paraphrase ensemble on five of six development models at 68 to 78 percent of its calls; on the held-out set it transferred in budget (79 to 87 percent), exceeding every fixed allocation on one model by a single question, losing one to two questions to the best fixed conditions on two, and seven to the paraphrase vote on one, so its measured value is the call saving; a sequential unanimity rule, a special case of early-stopping self-consistency, preserved self-consistency accuracy within one question everywhere at two to three calls. We release prompts, paraphrases, per-call predictions, and the scripts that recompute every table and statistic from the raw logs.

Declarations

CRediT authorship contribution statement

Chang Tan: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing — original draft, Writing — review & editing.

Declaration of competing interest

The author declares no competing interests. All model access was obtained through free-tier third-party endpoints; no provider had any role in the study design, the analysis, or the decision to submit.

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Data availability

All prompts, validated paraphrases with per-item validation verdicts, per-call raw predictions (with timestamps and token usage), analysis outputs, and the scripts that recompute every table and statistic directly from the raw logs are available at https://github.com/43aquarius/Test-TimeEnsemblingforLargeLanguageModels (releases v1.0.0–v1.8.0).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI writing assistants (large language model tools) to support drafting and editing of the manuscript text. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Appendix A. Greedy paraphrases: isolating the prompt-space effect

The standard paraphrase ensemble decodes each paraphrase once at $\tau = 0.7$, so its diversity mixes prompt-space variation with decoding stochasticity. The greedy-paraphrase control removes the second component by decoding each paraphrase at $\tau = 0$, yielding the noise-free ensemble PE^g-6 and a fragility indicator in which neither side involves sampling. Table A.1 reports the control for the five models whose provider quota allowed it.

The control separates three effects. First, for the flagship, degeneracy is not a temperature artifact: 98.9% of Qwen3.8-Max's questions return six identical answers even under greedy decoding of six different paraphrases, and the greedy paraphrase ensemble returns the same answer as the sampled paraphrase ensemble on 88 of 88 questions. The collapse is a property of the model-task pair, not of the sampling distribution around it. Second, for the mid-tier models, the prompt-space effect carries the paraphrase ensemble almost entirely: the PE^g-6 and PE-6 modal answers coincide on 31 of 32 questions for Agnes3.0-Flash and 38 of 39 for Agnes2.5-Flash, so the stochastic component of PE contributes almost nothing beyond the paraphrases

Table A.1: Greedy-paraphrase control. Accuracy (%) of the greedy baseline, SC-6, PE-6, and the greedy-decoded paraphrase ensemble PE^g-6; Frag.^{g} is the fraction of questions where SC-6 and PE^g-6 disagree (no decoding noise on the paraphrase side); Degen.^{g} is the fraction of questions where all six greedy paraphrase answers coincide.

Model	n	Greedy	SC-6	PE-6	PE^g-6	Ident.	Frag.^{g}	Degen.^{g}
Qwen3.8-Max	88	93.2	93.2	93.2	93.2	88/88	1.1	98.9
Qwen3.8-Omni	–	quota exhausted before the control could be run						
Agnes3.0-Flash	32	81.2	93.8	93.8	93.8	31/32	3.1	71.9
Agnes2.5-Flash	39	89.7	94.9	92.3	92.3	38/39	2.6	89.7
GPT-6-Luna	15	80.0	73.3	86.7	80.0	11/15	40.0	53.3
Gemini3.8-Flash	12	91.7	83.3	91.7	83.3	10/12	25.0	41.7

themselves in these cells. Third, for the prompt-brittle models, decoding noise and prompt variation interact: GPT-6-Luna's greedy paraphrase ensemble reaches 80.0% against 86.7% for the sampled version with modal-answer agreement on only 11 of 15 questions, and Gemini3.8-Flash 83.3% against 91.7% with agreement on 10 of 12, so for the most fragile models the sampled paraphrase ensemble retains a two-to-eight point edge over its noise-free counterpart, and the two ensembles disagree on a quarter of the questions. Fragility itself survives the control in strengthened form: GPT-6-Luna flips on 40.0% of questions and Gemini3.8-Flash on 25.0% when both sides are decoded greedily, which rules out sampling noise as the cause of paraphrase-induced flips and locates the instability in the model's response to the prompt's surface form.

Appendix B. Harder tasks: the MMLU-Pro probe

The flagship models saturate five of the six main-suite benchmarks, so the degenerate regime dominates their measurements. The MMLU-Pro probe asks whether a harder distribution restores answer entropy under the same direct-answering protocol. Table B.1 reports the probe; the coverage caveat is stated up front: on hard quantitative items the direct-answering instruction does not always suppress reasoning, and outputs truncated at the token cap lost extraction on a majority of Qwen3.8-Max's probe calls, leaving six complete questions (eight more with at least four parsed votes); the other models parsed cleanly.

The probe shows that difficulty alone does not determine the regime. GPT-6-Luna, the strongest model with full probe coverage, lands in the high-

Table B.1: MMLU-Pro probe: greedy accuracy, single-sample accuracy S1, and SC-6 accuracy, with mean support-normalized sample entropy, sample degeneracy rate, and the model's main-suite entropy for comparison.

Model	n	Greedy	S1	SC-6	Entropy	Degen.	Suite entropy
Qwen3.8-Max	6	50.0	50.0	50.0	0.153	83.3	0.018
Agnes3.0-Flash	8	12.5	6.2	0.0	0.937	0.0	0.162
Agnes2.5-Flash	13	84.6	76.9	84.6	0.241	69.2	0.055
GPT-6-Luna	15	66.7	66.7	73.3	0.261	66.7	0.267

entropy regime: greedy accuracy 66.7% with mean entropy 0.26 and a 66.7% degeneracy rate, and self-consistency lifts it to 73.3%; at $n = 15$ this 6.6 point difference is the movement of a single question and is reported as directional, not as evidence. Qwen3.8-Max remains essentially degenerate on the probe: its six complete questions are unanimous on five, and seven of the eight questions with at least four parsed votes return a single answer, so for this flagship a ten-option hard benchmark under direct answering still collapses to one answer per question. At the other end, Agnes3.0-Flash collapses into near-uniform guessing: entropy 0.94 with zero degeneracy, greedy accuracy 12.5%, and a self-consistency majority of 0% on its eight probe questions; at $n = 8$ these values are one to two questions from chance and are likewise directional, but they are consistent with the plurality-vote failure mode of Section 3.7 when the modal answer is wrong. Agnes2.5-Flash answers 84.6% correctly with partial degeneracy and no aggregation change. The probe is small, and we read it as directional evidence: harder tasks shift models along the entropy axis, but into which regime depends on the model, and the flagship's degeneracy is robust to at least this level of difficulty increase under direct answering.

Appendix C. Answering mode: direct answer versus chain of thought

The main suite answers multiple-choice questions directly, while self-consistency was originally motivated for chain-of-thought reasoning. The CoT condition therefore repeats the greedy baseline, the six-sample self-consistency, and, for stem-locus questions, the paraphrase ensemble under an explicit step-by-step instruction. Table C.1 reports the condition against the direct-answering cells on the same questions.

Table C.1: Chain-of-thought condition on multiple-choice questions: accuracy under greedy, SC-6, and PE-6 with CoT against the same conditions under direct answering on identical questions (Grd_{dir}, SC_{dir}, PE_{dir}), with mean answer entropy under both modes.

Model	n	Grd	SC-6	PE-6	Grd_{dir}	SC_{dir}	PE_{dir}	H	H_{dir}
Qwen3.8-Max	9	100.0	100.0	100.0	88.9	88.9	75.0	0.000	0.000
Agnes3.0-Flash	12	100.0	100.0	100.0	83.3	100.0	100.0	0.077	0.066
Agnes2.5-Flash	33	87.9	93.9	93.3	90.9	93.9	86.7	0.095	0.028
GPT-6-Luna	13	92.3	100.0	100.0	84.6	76.9	100.0	0.171	0.208

Chain-of-thought answering raises accuracy where it completes, but it does not dissolve the regime structure. For the flagship subset, CoT lifts Qwen3.8-Max from 88.9% to 100% on its nine completed questions while leaving the answer distribution fully degenerate: entropy stays at zero and all six samples agree on every question, so self-consistency remains vacuous. For the mid-tier models, CoT preserves or slightly enlarges diversity: Agnes2.5-Flash holds self-consistency at 93.9% in both modes on its 33 questions while greedy dips from 90.9% to 87.9% and entropy rises from 0.03 to 0.10, Agnes3.0-Flash gains from 83.3% to 100% under greedy on its 12 completed questions while remaining largely degenerate, and GPT-6-Luna's self-consistency improves from 76.9% to 100% on its 13 completed questions with its paraphrase ensemble at 100% in both modes. The pattern that survives the mode change is the important one: aggregation gains track the entropy regime in both modes, the degenerate stratum stays degenerate under CoT for the flagship, and the paraphrase side continues to carry signal only where entropy exists. Absolute accuracies are not comparable across modes, the CoT subsets are small, and the slow long-form generations bounded coverage for the rate-limited models, so we read this condition as a robustness check on the regime structure rather than as a mode comparison.

References

[1] M. Mizrahi, G. Kaplan, D. Malkin, R. Dror, D. Shahaf, G. Stanovsky, State of what art? a call for multi-prompt LLM evaluation, Transactions of the Association for Computational Linguistics 12 (2024) 933–949. doi:10.1162/tacl_a_00681.

[2] M. Sclar, Y. Choi, Y. Tsvetkov, A. Suhr, Quantifying language models' sensitivity to spurious features in prompt design, or: How i learned to

start worrying about prompt formatting, in: International Conference on Learning Representations (ICLR), 2024.

[3] A. Leidinger, R. van Rooij, E. Shutova, The language of prompting: What linguistic properties make a prompt successful?, in: Findings of the Association for Computational Linguistics: EMNLP 2023, 2023, pp. 9210–9232. doi:10.18653/v1/2023.findings-emnlp.618.

[4] K. Zhu, J. Wang, J. Zhou, Z. Wang, H. Chen, Y. Wang, L. Yang, W. Ye, Y. Zhang, N. Z. Gong, X. Xie, Promptrobust: Towards evaluating the robustness of large language models on adversarial prompts, arXiv preprint arXiv:2306.04528 (2023).

[5] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, D. Zhou, Self-consistency improves chain of thought reasoning in language models, in: International Conference on Learning Representations (ICLR), 2023.

[6] L. Kuhn, Y. Gal, S. Farquhar, Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation, in: International Conference on Learning Representations (ICLR), 2023.

[7] S. Farquhar, J. Kossen, L. Kuhn, Y. Gal, Detecting hallucinations in large language models using semantic entropy, Nature 630 (2024) 625–630. doi:10.1038/s41586-024-07421-0.

[8] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. V. Le, D. Zhou, Chain-of-thought prompting elicits reasoning in large language models, in: Advances in Neural Information Processing Systems, Vol. 35, 2022, pp. 24824–24837.

[9] P. Aggarwal, A. Madaan, Y. Yang, Mausam, Let's sample step by step: Adaptive-consistency for efficient reasoning and coding with LLMs, in: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, 2023, pp. 12375–12396. doi:10.18653/v1/2023.emnlp-main.761.

[10] Y. Li, P. Yuan, S. Feng, B. Pan, X. Wang, B. Sun, H. Wang, K. Li, Escape sky-high cost: Early-stopping self-consistency for multi-step reasoning, in: International Conference on Learning Representations (ICLR), 2024.

[11] X. Chen, R. Aksitov, U. Alon, J. Ren, K. Xiao, P. Yin, S. Prakash, C. Sutton, X. Wang, D. Zhou, Universal self-consistency for large language model generation, arXiv preprint arXiv:2311.17311 (2023).

[12] C. Snell, J. Lee, K. Xu, A. Kumar, Scaling LLM test-time compute optimally can be more effective than scaling parameters for reasoning, in: International Conference on Learning Representations (ICLR), 2025.

[13] B. Brown, J. Juravsky, R. Ehrlich, R. Clark, Q. V. Le, C. Ré, A. Mirhoseini, Large language monkeys: Scaling inference compute with repeated sampling, arXiv preprint arXiv:2407.21787 (2024).

[14] H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, K. Cobbe, Let's verify step by step, in: International Conference on Learning Representations (ICLR), 2024.

[15] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, I. Mordatch, Improving factuality and reasoning in language models through multiagent debate, in: Proceedings of the 41st International Conference on Machine Learning, Vol. 235 of Proceedings of Machine Learning Research, 2024, pp. 11733–11763.

[16] R. Naik, V. Chandrasekaran, M. Yuksekgonul, H. Palangi, B. Nushi, Diversity of thought improves reasoning abilities of LLMs, arXiv preprint arXiv:2310.07088 (2023).

[17] A. Smit, P. Duckworth, N. Grinsztajn, T. D. Barrett, A. Pretorius, Should we be going MAD? a look at multi-agent debate strategies for LLMs, arXiv preprint arXiv:2311.17371 (2023).

[18] S. Arora, A. Narayan, M. F. Chen, L. Orr, N. Guha, K. Bhatia, Z. Li, A. Guha, M. Lam, C. Ré, et al., Ask me anything: A simple strategy for prompting language models, arXiv preprint arXiv:2210.02441 (2022).

[19] S. Pitis, M. R. Zhang, A. Wang, J. Ba, Boosted prompt ensembles for large language models, arXiv preprint arXiv:2304.05970 (2023).

[20] K. Tian, E. Mitchell, A. Zhou, A. Sharma, R. Rafailov, H. Yao, C. Finn, C. D. Manning, Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback, in: Proceedings of the 2023 Conference on Empiri-

cal Methods in Natural Language Processing, 2023, pp. 5433–5442. doi:10.18653/v1/2023.emnlp-main.330.

[21] M. Xiong, Z. Hu, X. Lu, Y. Li, J. Fu, J. He, B. Hooi, Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs, in: International Conference on Learning Representations (ICLR), 2024.

[22] S. Kadavath, T. Conerly, A. Askell, T. Henighan, D. Drain, E. Perez, N. Schiefer, Z. Hatfield-Dodds, N. DasSarma, E. Tran-Johnson, et al., Language models (mostly) know what they know, arXiv preprint arXiv:2207.05221 (2022).

[23] X. Liu, T. Chen, L. Da, C. Chen, Z. Lin, H. Wei, Uncertainty quantification and confidence calibration in large language models: A survey, arXiv preprint arXiv:2503.15850 (2025).

[24] X. Gao, J. Zhang, L. Mouatadid, K. Das, SPUQ: Perturbation-based uncertainty quantification for large language models, in: Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2024, pp. 2336–2346. doi:10.18653/v1/2024.eacl-long.143.

[25] B. Hou, Y. Liu, K. Qian, J. Andreas, S. Chang, Y. Zhang, Decomposing uncertainty for large language models through input clarification ensembling, in: Proceedings of the 41st International Conference on Machine Learning, Vol. 235 of Proceedings of Machine Learning Research, 2024, pp. 19023–19042.

[26] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: Proceedings of the 34th International Conference on Machine Learning, 2017, pp. 1321–1330.

[27] T. Lanham, A. Chen, A. Radhakrishnan, B. Steiner, C. Denison, D. Hernandez, D. Li, E. Durmus, E. Hubinger, J. Kernion, et al., Measuring faithfulness in chain-of-thought reasoning, arXiv preprint arXiv:2307.13702 (2023).

[28] M. d. C. de Caritat, Marie Jean Antoine Nicolas, Essai sur l'application de l'analyse à la probabilité des décisions rendues à la pluralité des voix, Imprimerie Royale, Paris, 1785.

[29] S. Lin, J. Hilton, O. Evans, Truthfulqa: Measuring how models mimic human falsehoods, in: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics, 2022, pp. 3214–3252. doi:10.18653/v1/2022.acl-long.229.

[30] Y. Bisk, R. Zellers, R. Le Bras, J. Gao, Y. Choi, PIQA: Reasoning about physical commonsense in natural language, in: Proceedings of the Thirty-Fourth AAAI Conference on Artificial Intelligence, 2020, pp. 7432–7439. doi:10.1609/aaai.v34i05.6239.

[31] J. Liu, L. Cui, H. Liu, D. Huang, Y. Wang, Y. Zhang, Logiqa: A challenge dataset for machine reading comprehension with logical reasoning, in: Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI), 2020, pp. 3622–3628. doi:10.24963/ijcai.2020/501.

[32] W. Zhong, R. Cui, Y. Guo, Y. Liang, S. Lu, Y. Wang, A. Saied, W. Chen, N. Duan, AGIEval: A human-centric benchmark for evaluating foundation models, in: Findings of the Association for Computational Linguistics: NAACL 2024, 2024, pp. 2299–2314. doi:10.18653/v1/2024.findings-naacl.149.

[33] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, J. Schulman, Training verifiers to solve math word problems, arXiv preprint arXiv:2110.14168 (2021).

[34] A. Yang, et al., Qwen3 technical report, arXiv preprint arXiv:2505.09388 (2025).

[35] Q. Team, Qwen technical report, arXiv preprint arXiv:2309.16609 (2023).

[36] Agnes AI, Agnes ai api documentation, https://agnes-ai.com, accessed October 2026 (2026).

[37] OpenAI, Gpt-4 technical report, arXiv preprint arXiv:2303.08774 (2023).

[38] OpenAI, Gpt-5 system card, Tech. rep., OpenAI (2025).
URL https://openai.com/index/gpt-5-system-card

[39] G. Comanici, et al., Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context and next generation agentic capabilities, arXiv preprint arXiv:2507.06261 (2025).

[40] DeepSeek-AI, DeepSeek API Documentation, https://api-docs.deepseek.com, accessed October 2026 via third-party providers (Aliyun MaaS, Apinex) (2026).

[41] Y. Wang, X. Ma, G. Zhang, Y. Ni, A. Chandra, S. Guo, W. Ren, A. Arulraj, X. He, Z. Jiang, T. Li, M. Ku, K. Wang, A. Zhuang, R. Fan, X. Yue, W. Chen, MMLU-Pro: A more robust and challenging multi-task language understanding benchmark, in: Advances in Neural Information Processing Systems, Vol. 37, 2024.

[42] A. Holtzman, P. West, V. Shwartz, Y. Choi, L. Zettlemoyer, Surface form competition: Why the highest probability answer isn't always right, in: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, 2021, pp. 7038–7051. doi:10.18653/v1/2021.emnlp-main.564.